

Código Fuente - ElotApp

Archivo: java\com\example\proyectofinalappmobile\ai\AiCallback.java

```
``java
```

```
package com.example.proyectofinalappmobile.ai;
```

```
import com.example.proyectofinalappmobile.data.models.ChatIntent;
```

```
public interface AiCallback {  
    void onResult(ChatIntent intent);  
    void onError(Exception e);  
}
```

```
``
```

Archivo: java\com\example\proyectofinalappmobile\ai\AiClient.java

```
``java
```

```
package com.example.proyectofinalappmobile.ai;
```

```
public interface AiClient {  
    void ask(String question, AiCallback callback);  
}
```

```
``
```

Archivo: java\com\example\proyectofinalappmobile\ai\DummyAiClient.java

```
``java
```

```
package com.example.proyectofinalappmobile.ai;
```

```
import com.example.proyectofinalappmobile.data.models.ChatIntent;
```

```
/**
```

```
 * Cliente de IA dummy para pruebas.
```

```
 * Analiza preguntas simples y retorna ChatIntent basado en palabras clave.
```

```
*/
```

```

public class DummyAiClient implements AiClient {

    @Override
    public void ask(String question, AiCallback callback) {
        // Simular procesamiento asíncrono
        new Thread(() -> {
            try {
                Thread.sleep(500); // Simular latencia de red

                ChatIntent intent = analyzeQuestion(question);
                callback.onResult(intent);

            } catch (Exception e) {
                callback.onError(e);
            }
        }).start();
    }

    private ChatIntent analyzeQuestion(String question) {
        String q = question.toLowerCase();

        // Detectar intención de stock
        if (q.contains("cuánto") || q.contains("stock") || q.contains("tengo")) {
            String productName = extractProductName(q);
            return new ChatIntent(ChatIntent.IntentType.GET_STOCK, productName);
        }

        // Detectar intención de ventas
        if (q.contains("vend") || q.contains("venta") || q.contains("semana")) {
            String productName = extractProductName(q);
            return new ChatIntent(ChatIntent.IntentType.GET_WEEKLY_SALES, productName,
0);
        }

        // Detectar intención de stock bajo
        if (q.contains("bajo") || q.contains("mínimo") || q.contains("falta")) {
            return new ChatIntent(ChatIntent.IntentType.GET_LOW_STOCK);
        }

        return new ChatIntent(ChatIntent.IntentType.UNKNOWN);
    }

    private String extractProductName(String question) {
        // Buscar palabras clave de productos comunes
        if (question.contains("elote")) {
            return "elotes";
        }
        if (question.contains("esquite")) {

```

```

        return "esquites";
    }
    if (question.contains("mayo")) {
        return "mayonesa";
    }
    if (question.contains("queso")) {
        return "queso";
    }
    if (question.contains("chile")) {
        return "chile";
    }
    return "producto";
}
}

```

``

Archivo: java\com\example\proyectofinalappmobile\ai\GeminiAiClient.java

``java

```
package com.example.proyectofinalappmobile.ai;
```

```
import android.content.Context;
import android.database.Cursor;
```

```
import com.example.proyectofinalappmobile.data.dao.ProductDao;
import com.example.proyectofinalappmobile.data.dao.MovementDao;
import com.example.proyectofinalappmobile.data.models.ChatIntent;
import com.google.gson.Gson;
import com.google.gson.JsonObject;
import com.google.gson.JsonArray;
```

```
import java.io.IOException;
```

```
import okhttp3.Call;
import okhttp3.Callback;
import okhttp3.MediaType;
import okhttp3.OkHttpClient;
import okhttp3.Request;
import okhttp3.RequestBody;
import okhttp3.Response;
```

```
/**
```

```
 * Cliente que conecta con la API de Gemini para analizar preguntas en lenguaje natural
```

```
* y convertirlas en intenciones estructuradas.  
*/
```

```
public class GeminiAiClient implements AiClient {
```

```
    private final OkHttpClient client;  
    private final Gson gson;  
    private final Context context;  
    private final ProductDao productDao;  
    private final MovementDao movementDao;
```

```
    public GeminiAiClient(Context context) {  
        this.context = context;  
        this.client = new OkHttpClient();  
        this.gson = new Gson();  
        this.productDao = new ProductDao(context);  
        this.movementDao = new MovementDao(context);  
    }
```

```
@Override
```

```
public void ask(String question, AiCallback callback) {  
    // Verificar si estÃ¡ configurada la API key  
    if (!GeminiConfig.isConfigured()) {  
        // Si no hay API key, usar anÃ¡lisis bÃ¡sico  
        android.util.Log.d("GeminiAiClient", "API key no configurada, usando fallback");  
        ChatIntent fallbackIntent = analyzeFallback(question);  
        callback.onResult(fallbackIntent);  
        return;  
    }
```

```
    android.util.Log.d("GeminiAiClient", "Enviando pregunta a Gemini: " + question);
```

```
    // Obtener datos del inventario
```

```
    String inventoryData = getInventoryData();  
    android.util.Log.d("GeminiAiClient", "Datos de inventario: " + inventoryData);
```

```
    // Crear el prompt para Gemini con contexto completo
```

```
    String systemPrompt = "Eres un asistente personal para la DUEÃ±A de un negocio  
llamda ariana que es de venta de elotes.\n\n" +  
        "La dueÃ±a del negocio tiene estos productos en su inventario:\n" + inventoryData  
    + "\n\n" +  
        "La dueÃ±a te pregunta: \"" + question + "\"\n\n" +  
        "IMPORTANTE: TÃº NO eres un cliente. Eres el ASISTENTE de la dueÃ±a del  
negocio. " +  
        "Responde como si estuvieras ayudando a la dueÃ±a a gestionar su negocio. " +  
        "Usa emojis cuando sea apropiado. " +  
        "Si preguntan por stock, ventas o inventario, consulta los datos de arriba y  
responde basÃ¡ndote en ellos. " +
```

```

+         "Si es una pregunta general del negocio, da consejos Ãtiles para el negocio.\n\n"

        "Tu respuesta:";

// Crear el request body en formato Gemini
JsonObject requestJson = new JsonObject();
JsonArray contents = new JsonArray();
JsonObject content = new JsonObject();
JsonArray parts = new JsonArray();
JsonObject part = new JsonObject();

part.addProperty("text", systemPrompt);
parts.add(part);
content.add("parts", parts);
contents.add(content);
requestJson.add("contents", contents);

String jsonBody = gson.toJson(requestJson);

RequestBody body = RequestBody.create(
    jsonBody,
    MediaType.parse("application/json")
);

Request request = new Request.Builder()
    .url(GeminiConfig.getApiUrl())
    .post(body)
    .build();

client.newCall(request).enqueue(new Callback() {
    @Override
    public void onFailure(Call call, IOException e) {
        // Si falla la API, usar anÃlisis bÃsico
        android.util.Log.e("GeminiAiClient", "Error en llamada a Gemini: " +
e.getMessage());
        ChatIntent fallbackIntent = analyzeFallback(question);
        callback.onResult(fallbackIntent);
    }

    @Override
    public void onResponse(Call call, Response response) throws IOException {
        android.util.Log.d("GeminiAiClient", "Respuesta recibida. CÃdigo: " +
response.code());

        if (!response.isSuccessful()) {
            android.util.Log.e("GeminiAiClient", "Respuesta no exitosa: " +
response.message());
            // Si hay error, usar anÃlisis bÃsico

```

```

        ChatIntent fallbackIntent = analyzeFallback(question);
        callback.onResult(fallbackIntent);
        return;
    }

    try {
        String responseBody = response.body().string();
        android.util.Log.d("GeminiAiClient", "Respuesta de Gemini: " +
responseBody.substring(0, Math.min(200, responseBody.length())));

        JsonObject jsonResponse = gson.fromJson(responseBody, JsonObject.class);

        // Extraer el texto de la respuesta de Gemini
        String aiResponse = jsonResponse
            .getAsJsonArray("candidates")
            .get(0).getAsJsonObject()
            .getAsJsonObject("content")
            .getAsJsonArray("parts")
            .get(0).getAsJsonObject()
            .get("text").getString();

        // Crear un intent DIRECT_ANSWER con la respuesta de Gemini
        ChatIntent intent = new ChatIntent(ChatIntent.IntentType.DIRECT_ANSWER);
        intent.setDirectAnswer(aiResponse);
        android.util.Log.d("GeminiAiClient", "Respuesta parseada correctamente");
        callback.onResult(intent);

    } catch (Exception e) {
        // Si falla el parsing, usar análisis básico
        android.util.Log.e("GeminiAiClient", "Error parseando respuesta: " +
e.getMessage());
        ChatIntent fallbackIntent = analyzeFallback(question);
        callback.onResult(fallbackIntent);
    }
}

private String getInventoryData() {
    StringBuilder data = new StringBuilder();

    // Obtener todos los productos
    Cursor products = productDao.getAll();
    if (products != null && products.moveToFirst()) {
        data.append("PRODUCTOS EN INVENTARIO:\n");
        do {
            String name = products.getString(products.getColumnIndexOrThrow("name"));
            int stock = products.getInt(products.getColumnIndexOrThrow("stock"));

```

```

        int minStock = products.getInt(products.getColumnIndexOrThrow("min_stock"));
        double price = products.getDouble(products.getColumnIndexOrThrow("price"));

        data.append("- ").append(name).append(": ")
            .append(stock).append(" unidades ")
            .append("(mÁnimo: ").append(minStock).append(", ")
            .append("precio: $").append(price).append("\n");
    } while (products.moveToNext());
    products.close();
}

// Obtener ventas de la semana
int weeklySales = movementDao.getWeeklySalesCount();
double weeklyRevenue = movementDao.getWeeklyRevenue();

data.append("\nVENTAS ESTA SEMANA:\n");
data.append("- Total de ventas: ").append(weeklySales).append("\n");
data.append("- Ingresos: $").append(String.format("%.2f",
weeklyRevenue)).append("\n");

return data.toString();
}

// Análisis básico por palabras clave como fallback
private ChatIntent analyzeFallback(String question) {
    String q = question.toLowerCase();

    if (q.contains("cuÁnto") || q.contains("stock") || q.contains("tengo")) {
        String productName = extractProductName(q);
        return new ChatIntent(ChatIntent.IntentType.GET_STOCK, productName);
    }

    if (q.contains("vend") || q.contains("venta") || q.contains("semana")) {
        String productName = extractProductName(q);
        return new ChatIntent(ChatIntent.IntentType.GET_WEEKLY_SALES, productName,
0);
    }

    if (q.contains("bajo") || q.contains("mÁnimo") || q.contains("falta")) {
        return new ChatIntent(ChatIntent.IntentType.GET_LOW_STOCK);
    }

    return new ChatIntent(ChatIntent.IntentType.UNKNOWN);
}

private String extractProductName(String question) {
    if (question.contains("elote")) return "elotes";
    if (question.contains("esquite")) return "esquites";
}

```

```

        if (question.contains("mayo")) return "mayonesa";
        if (question.contains("queso")) return "queso";
        if (question.contains("chile")) return "chile";
        return "producto";
    }
}

```

``

Archivo: java\com\example\proyectofinalappmobile\ai\GeminiClient.java

``java

```
package com.example.proyectofinalappmobile.ai;
```

```
import com.example.proyectofinalappmobile.data.models.ChatIntent;
```

```
/**
```

```

 * Cliente para Google Gemini API.
 * TODO: Implementar con Retrofit cuando se tenga API key.
 *
 * Referencias:
 * - https://ai.google.dev/docs
 * - Function Calling: https://ai.google.dev/docs/function\_calling
 */

```

```
public class GeminiClient implements AiClient {
```

```

    private static final String API_KEY =
    "AlzaSyDeHYe7zUHyT9ZE_w5GjG6ATnWCExxPIZI";
    private static final String API_URL =
    "https://generativelanguage.googleapis.com/v1beta/models/gemini-pro:generateContent";

```

```
@Override
```

```

public void ask(String question, AiCallback callback) {
    // TODO: Implementar llamada real a Gemini API
    // Por ahora, usar DummyAiClient como fallback
    new DummyAiClient().ask(question, callback);
}

```

```
/*
```

```

 * Ejemplo de implementación con Retrofit:
 *
 * GeminiRequest request = new GeminiRequest(question);
 *
 * apiService.generateContent(API_KEY, request)
 *     .enqueue(new Callback<GeminiResponse>() {

```



```

    *      @Override
    *      public void onResponse(Call<GeminiResponse> call,
Response<GeminiResponse> response) {
    *          if (response.isSuccessful() && response.body() != null) {
    *              ChatIntent intent = parseResponse(response.body());
    *              callback.onResult(intent);
    *          } else {
    *              callback.onError(new Exception("Error en la respuesta de la API"));
    *          }
    *      }
    *
    *      @Override
    *      public void onFailure(Call<GeminiResponse> call, Throwable t) {
    *          callback.onError(new Exception(t));
    *      }
    *  });
    */
}

```

```

private ChatIntent parseResponse(Object response) {
    // TODO: Parsear respuesta de Gemini y extraer intent
    return new ChatIntent(ChatIntent.IntentType.UNKNOWN);
}
}

```

``

Archivo: java\com\example\proyectofinalappmobile\ai\GeminiConfig.java

``java

```
package com.example.proyectofinalappmobile.ai;
```

```
/**
```

```
* Configuraci3n de la API de Gemini.
```

```
* IMPORTANTE: Consigue tu API key gratis en https://makersuite.google.com/app/apikey
```

```
*/
```

```
public class GeminiConfig {
```

```
    // 8Y” COLOCA TU API KEY AQUA
```

```
    // Para obtenerla:
```

```
    // 1. Ve a https://makersuite.google.com/app/apikey
```

```
    // 2. Inicia sesi3n con tu cuenta de Google
```

```
    // 3. Haz clic en "Create API Key"
```

```
    // 4. Copia la key y p3gala abajo entre las comillas
```

```

    public static final String API_KEY = "AlzaSyDeHYe7zUHyT9ZE_w5GjG6ATnWCExxPIZI";

    // URL del endpoint de Gemini
    public static final String getApiUrl() {
        return
"https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash-lite:generateCo
ntent?key=" + API_KEY;
    }

    // Verificar si la API key está configurada
    public static boolean isConfigured() {
        return API_KEY != null && !API_KEY.equals("TU_API_KEY_AQUI") &&
!API_KEY.isEmpty();
    }
}

```

``

Archivo: java\com\example\proyectofinalappmobile\ai\OpenAiClient.java

``java

```

package com.example.proyectofinalappmobile.ai;

import com.example.proyectofinalappmobile.data.models.ChatIntent;

/**
 * Cliente para OpenAI GPT API.
 * TODO: Implementar con Retrofit cuando se tenga API key.
 *
 * Referencias:
 * - https://platform.openai.com/docs/api-reference
 * - Function Calling: https://platform.openai.com/docs/guides/function-calling
 */
public class OpenAiClient implements AiClient {

    private static final String API_KEY = "YOUR_OPENAI_API_KEY";
    private static final String API_URL = "https://api.openai.com/v1/chat/completions";

    @Override
    public void ask(String question, AiCallback callback) {
        // TODO: Implementar llamada real a OpenAI API
        // Por ahora, usar DummyAiClient como fallback
        new DummyAiClient().ask(question, callback);
    }
}

```

```

/*
 * Ejemplo de implementación con Retrofit:
 *
 * OpenAiRequest request = new OpenAiRequest.Builder()
 *     .model("gpt-4")
 *     .addMessage("user", question)
 *     .addFunction("get_stock", "Obtiene el stock de un producto", getStockSchema())
 *     .addFunction("get_sales", "Obtiene ventas de un producto", getSalesSchema())
 *     .addFunction("get_low_stock", "Lista productos con stock bajo",
getLowStockSchema())
 *     .build();
 *
 * apiService.chatCompletion("Bearer " + API_KEY, request)
 *     .enqueue(new Callback<OpenAiResponse>() {
 *         @Override
 *         public void onResponse(Call<OpenAiResponse> call,
Response<OpenAiResponse> response) {
 *             if (response.isSuccessful() && response.body() != null) {
 *                 ChatIntent intent = parseResponse(response.body());
 *                 callback.onResult(intent);
 *             } else {
 *                 callback.onError(new Exception("Error en la respuesta de la API"));
 *             }
 *         }
 *     });
 *
 * @Override
 * public void onFailure(Call<OpenAiResponse> call, Throwable t) {
 *     callback.onError(new Exception(t));
 * }
 * });
 */
}

private ChatIntent parseResponse(Object response) {
    // TODO: Parsear respuesta de OpenAI y extraer function call
    return new ChatIntent(ChatIntent.IntentType.UNKNOWN);
}
}

```

``

Archivo: java\com\example\proyectofinalappmobile\data\dao\MovementDao.java

``java

```

package com.example.proyectofinalappmobile.data.dao;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;

import com.example.proyectofinalappmobile.data.db.ElotDbHelper;
import com.example.proyectofinalappmobile.data.models.Movement;

public class MovementDao {

    private final ElotDbHelper dbHelper;

    public MovementDao(Context context) {
        dbHelper = new ElotDbHelper(context);
    }

    public long insert(Movement m) {
        SQLiteDatabase db = dbHelper.getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put("product_id", m.getProductId());
        values.put("type", m.getType());
        values.put("qty", m.getQty());
        values.put("date", m.getDate());
        return db.insert("movements", null, values);
    }

    // ventas de un periodo simple (ej. Ãltima semana)
    public int getSalesForLastDays(long productId, int days) {
        long now = System.currentTimeMillis();
        long from = now - days * 24L * 60L * 60L * 1000L;

        SQLiteDatabase db = dbHelper.getReadableDatabase();
        Cursor c = db.rawQuery(
            "SELECT SUM(qty) FROM movements " +
            "WHERE product_id=? AND type='SALE' AND date >= ?",
            new String[]{ String.valueOf(productId), String.valueOf(from) }
        );
        int total = 0;
        if (c.moveToFirst()) {
            total = c.getInt(0);
        }
        c.close();
        return total;
    }
}

```

```

public Cursor getAll() {
    SQLiteDatabase db = dbHelper.getReadableDatabase();
    return db.query("movements", null, null, null, null, null, "date DESC");
}

```

```

public Cursor getByProductId(long productId) {
    SQLiteDatabase db = dbHelper.getReadableDatabase();
    return db.query(
        "movements",
        null,
        "product_id=?",
        new String[]{String.valueOf(productId)},
        null, null,
        "date DESC"
    );
}

```

// Obtener ventas de esta semana

```

public int getWeeklySalesCount() {
    long now = System.currentTimeMillis();
    long weekAgo = now - 7L * 24L * 60L * 60L * 1000L;

    SQLiteDatabase db = dbHelper.getReadableDatabase();
    Cursor c = db.rawQuery(
        "SELECT COUNT(*) FROM movements WHERE type='SALE' AND date >= ?",
        new String[]{String.valueOf(weekAgo)}
    );
    int count = 0;
    if (c.moveToFirst()) {
        count = c.getInt(0);
    }
    c.close();
    return count;
}

```

// Obtener ventas desde una fecha específica (para semana manual)

```

public int getCustomWeeklySalesCount(long startDate) {
    SQLiteDatabase db = dbHelper.getReadableDatabase();
    Cursor c = db.rawQuery(
        "SELECT COUNT(*) FROM movements WHERE type='SALE' AND date >= ?",
        new String[]{String.valueOf(startDate)}
    );
    int count = 0;
    if (c.moveToFirst()) {
        count = c.getInt(0);
    }
    c.close();
    return count;
}

```

```
}
```

```
// Obtener ingresos de esta semana
```

```
public double getWeeklyRevenue() {
```

```
    long now = System.currentTimeMillis();
```

```
    long weekAgo = now - 7L * 24L * 60L * 60L * 1000L;
```

```
    SQLiteDatabase db = dbHelper.getReadableDatabase();
```

```
    Cursor c = db.rawQuery(
```

```
        "SELECT SUM(m.qty * p.price) FROM movements m " +
```

```
        "JOIN products p ON m.product_id = p.id " +
```

```
        "WHERE m.type='SALE' AND m.date >= ?",
```

```
        new String[]{String.valueOf(weekAgo)}
```

```
    );
```

```
    double revenue = 0.0;
```

```
    if (c.moveToFirst() && !c.isNull(0)) {
```

```
        revenue = c.getDouble(0);
```

```
    }
```

```
    c.close();
```

```
    return revenue;
```

```
}
```

```
// Obtener ingresos desde una fecha específica (para semana manual)
```

```
public double getCustomWeeklyRevenue(long startDate) {
```

```
    SQLiteDatabase db = dbHelper.getReadableDatabase();
```

```
    Cursor c = db.rawQuery(
```

```
        "SELECT SUM(m.qty * p.price) FROM movements m " +
```

```
        "JOIN products p ON m.product_id = p.id " +
```

```
        "WHERE m.type='SALE' AND m.date >= ?",
```

```
        new String[]{String.valueOf(startDate)}
```

```
    );
```

```
    double revenue = 0.0;
```

```
    if (c.moveToFirst() && !c.isNull(0)) {
```

```
        revenue = c.getDouble(0);
```

```
    }
```

```
    c.close();
```

```
    return revenue;
```

```
}
```

```
// Obtener total de movimientos
```

```
public int getTotalMovementsCount() {
```

```
    SQLiteDatabase db = dbHelper.getReadableDatabase();
```

```
    Cursor c = db.rawQuery("SELECT COUNT(*) FROM movements", null);
```

```
    int count = 0;
```

```
    if (c.moveToFirst()) {
```

```
        count = c.getInt(0);
```

```
    }
```

```
    c.close();
```

```

        return count;
    }

    // Obtener todos los movimientos con nombres de productos
    public Cursor getAllWithProductNames() {
        SQLiteDatabase db = dbHelper.getReadableDatabase();
        return db.rawQuery(
            "SELECT m.id AS _id, " +
            "    CASE m.type " +
            "        WHEN 'SALE' THEN 'ðŸ’œ' " +
            "        WHEN 'IN' THEN 'ðŸ’ˆ' " +
            "        WHEN 'OUT' THEN 'ðŸ’˜' " +
            "        ELSE 'ðŸ’ˆ' " +
            "    END AS type_icon, " +
            "    p.name AS product_name, " +
            "    ('Cantidad: ' || m.qty || ' ' || p.unit) AS quantity_text, " +
            "    datetime(m.date/1000, 'unixepoch', 'localtime') AS date_text " +
            "FROM movements m " +
            "JOIN products p ON m.product_id = p.id " +
            "ORDER BY m.date DESC",
            null
        );
    }
}

```

```

    // Obtener movimientos de esta semana
    public Cursor getWeeklyMovementsWithProductNames() {
        long now = System.currentTimeMillis();
        long weekAgo = now - 7L * 24L * 60L * 60L * 1000L;

        SQLiteDatabase db = dbHelper.getReadableDatabase();
        return db.rawQuery(
            "SELECT m.id AS _id, " +
            "    CASE m.type " +
            "        WHEN 'SALE' THEN 'ðŸ’œ' " +
            "        WHEN 'IN' THEN 'ðŸ’ˆ' " +
            "        WHEN 'OUT' THEN 'ðŸ’˜' " +
            "        ELSE 'ðŸ’ˆ' " +
            "    END AS type_icon, " +
            "    p.name AS product_name, " +
            "    ('Cantidad: ' || m.qty || ' ' || p.unit) AS quantity_text, " +
            "    datetime(m.date/1000, 'unixepoch', 'localtime') AS date_text " +
            "FROM movements m " +
            "JOIN products p ON m.product_id = p.id " +
            "WHERE m.date >= ? " +
            "ORDER BY m.date DESC",
            new String[]{String.valueOf(weekAgo)}
        );
    }
}

```

```

// Obtener movimientos desde una fecha específica (para semana manual)
public Cursor getCustomWeeklyMovementsWithProductNames(long startDate) {
    SQLiteDatabase db = dbHelper.getReadableDatabase();
    return db.rawQuery(
        "SELECT m.id AS _id, " +
        "    CASE m.type " +
        "        WHEN 'SALE' THEN '📉' " +
        "        WHEN 'IN' THEN '📈' " +
        "        WHEN 'OUT' THEN '📊' " +
        "        ELSE '📋' " +
        "    END AS type_icon, " +
        "    p.name AS product_name, " +
        "    ('Cantidad: ' || m.qty || ' ' || p.unit) AS quantity_text, " +
        "    datetime(m.date/1000, 'unixepoch', 'localtime') AS date_text " +
        "FROM movements m " +
        "JOIN products p ON m.product_id = p.id " +
        "WHERE m.date >= ? " +
        "ORDER BY m.date DESC",
        new String[]{String.valueOf(startDate)}
    );
}

```

```

// Obtener movimientos de este mes
public Cursor getMonthlyMovementsWithProductNames() {
    long now = System.currentTimeMillis();
    long monthAgo = now - 30L * 24L * 60L * 60L * 1000L;

    SQLiteDatabase db = dbHelper.getReadableDatabase();
    return db.rawQuery(
        "SELECT m.id AS _id, " +
        "    CASE m.type " +
        "        WHEN 'SALE' THEN '📉' " +
        "        WHEN 'IN' THEN '📈' " +
        "        WHEN 'OUT' THEN '📊' " +
        "        ELSE '📋' " +
        "    END AS type_icon, " +
        "    p.name AS product_name, " +
        "    ('Cantidad: ' || m.qty || ' ' || p.unit) AS quantity_text, " +
        "    datetime(m.date/1000, 'unixepoch', 'localtime') AS date_text " +
        "FROM movements m " +
        "JOIN products p ON m.product_id = p.id " +
        "WHERE m.date >= ? " +
        "ORDER BY m.date DESC",
        new String[]{String.valueOf(monthAgo)}
    );
}

```



```

// Obtener ventas del mes
public int getMonthlySalesCount() {
    long now = System.currentTimeMillis();
    long monthAgo = now - 30L * 24L * 60L * 60L * 1000L;

    SQLiteDatabase db = dbHelper.getReadableDatabase();
    Cursor c = db.rawQuery(
        "SELECT COUNT(*) FROM movements WHERE type='SALE' AND date >= ?",
        new String[]{String.valueOf(monthAgo)}
    );
    int count = 0;
    if (c.moveToFirst()) {
        count = c.getInt(0);
    }
    c.close();
    return count;
}

```

```

// Obtener ingresos del mes
public double getMonthlyRevenue() {
    long now = System.currentTimeMillis();
    long monthAgo = now - 30L * 24L * 60L * 60L * 1000L;

    SQLiteDatabase db = dbHelper.getReadableDatabase();
    Cursor c = db.rawQuery(
        "SELECT SUM(m.qty * p.price) FROM movements m " +
        "JOIN products p ON m.product_id = p.id " +
        "WHERE m.type='SALE' AND m.date >= ?",
        new String[]{String.valueOf(monthAgo)}
    );
    double revenue = 0.0;
    if (c.moveToFirst() && !c.isNull(0)) {
        revenue = c.getDouble(0);
    }
    c.close();
    return revenue;
}

```

```

// Obtener ventas totales
public int getTotalSalesCount() {
    SQLiteDatabase db = dbHelper.getReadableDatabase();
    Cursor c = db.rawQuery(
        "SELECT COUNT(*) FROM movements WHERE type='SALE'", null
    );
    int count = 0;
    if (c.moveToFirst()) {
        count = c.getInt(0);
    }
}

```

```

        c.close();
        return count;
    }

    // Obtener ingresos totales
    public double getTotalRevenue() {
        SQLiteDatabase db = dbHelper.getReadableDatabase();
        Cursor c = db.rawQuery(
            "SELECT SUM(m.qty * p.price) FROM movements m " +
            "JOIN products p ON m.product_id = p.id " +
            "WHERE m.type='SALE'", null
        );
        double revenue = 0.0;
        if (c.moveToFirst() && !c.isNull(0)) {
            revenue = c.getDouble(0);
        }
        c.close();
        return revenue;
    }

```

```

    // Obtener un movimiento por ID
    public Movement getById(long movementId) {
        SQLiteDatabase db = dbHelper.getReadableDatabase();
        Cursor c = db.query(
            "movements",
            null,
            "id=?",
            new String[]{String.valueOf(movementId)},
            null, null, null
        );
        Movement movement = null;
        if (c.moveToFirst()) {
            movement = new Movement(
                c.getLong(c.getColumnIndexOrThrow("id")),
                c.getLong(c.getColumnIndexOrThrow("product_id")),
                c.getString(c.getColumnIndexOrThrow("type")),
                c.getInt(c.getColumnIndexOrThrow("qty")),
                c.getLong(c.getColumnIndexOrThrow("date"))
            );
        }
        c.close();
        return movement;
    }

```

```

    // Actualizar un movimiento
    public int update(Movement movement) {
        SQLiteDatabase db = dbHelper.getWritableDatabase();
        ContentValues values = new ContentValues();
    }

```

```

        values.put("product_id", movement.getProductId());
        values.put("type", movement.getType());
        values.put("qty", movement.getQty());
        values.put("date", movement.getDate());
        return db.update("movements", values, "id=?", new
String[]{String.valueOf(movement.getId())});
    }

    // Eliminar un movimiento
    public int delete(long movementId) {
        SQLiteDatabase db = dbHelper.getWritableDatabase();
        return db.delete("movements", "id=?", new String[]{String.valueOf(movementId)});
    }

    // Obtener productos más vendidos (top sellers)
    public Cursor getTopSellingProducts(int limit) {
        SQLiteDatabase db = dbHelper.getReadableDatabase();
        return db.rawQuery(
            "SELECT p.id AS _id, " +
            "    p.name, " +
            "    SUM(m.qty) AS total_sold, " +
            "    SUM(m.qty * p.price) AS total_revenue, " +
            "    COUNT(m.id) AS times_sold, " +
            "    p.unit " +
            "FROM movements m " +
            "JOIN products p ON m.product_id = p.id " +
            "WHERE m.type = 'SALE' " +
            "GROUP BY p.id " +
            "ORDER BY total_sold DESC " +
            "LIMIT ?",
            new String[]{String.valueOf(limit)}
        );
    }

    // Obtener productos más vendidos de esta semana
    public Cursor getTopSellingProductsThisWeek(int limit, long startDate) {
        SQLiteDatabase db = dbHelper.getReadableDatabase();
        return db.rawQuery(
            "SELECT p.id AS _id, " +
            "    p.name, " +
            "    SUM(m.qty) AS total_sold, " +
            "    SUM(m.qty * p.price) AS total_revenue, " +
            "    COUNT(m.id) AS times_sold, " +
            "    p.unit " +
            "FROM movements m " +
            "JOIN products p ON m.product_id = p.id " +
            "WHERE m.type = 'SALE' AND m.date >= ? " +
            "GROUP BY p.id " +

```

```

        "ORDER BY total_sold DESC " +
        "LIMIT ?",
        new String[]{String.valueOf(startDate), String.valueOf(limit)}
    );
}

// Obtener ventas por día (últimos 7 días)
public Cursor getDailySalesStats(int days) {
    SQLiteDatabase db = dbHelper.getReadableDatabase();
    long now = System.currentTimeMillis();
    long from = now - days * 24L * 60L * 60L * 1000L;

    return db.rawQuery(
        "SELECT date(m.date/1000, 'unixepoch', 'localtime') AS day, " +
        "    COUNT(*) AS sales_count, " +
        "    SUM(m.qty) AS items_sold, " +
        "    SUM(m.qty * p.price) AS revenue " +
        "FROM movements m " +
        "JOIN products p ON m.product_id = p.id " +
        "WHERE m.type = 'SALE' AND m.date >= ? " +
        "GROUP BY day " +
        "ORDER BY day DESC",
        new String[]{String.valueOf(from)}
    );
}
}

```

..

Archivo: java\com\example\proyectofinalappmobile\data\dao\ProductDao.java

``java

```
package com.example.proyectofinalappmobile.data.dao;
```

```
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
```

```
import com.example.proyectofinalappmobile.data.db.ElotDbHelper;
import com.example.proyectofinalappmobile.data.models.Product;
```

```
public class ProductDao {
```

```

private final ElotDbHelper dbHelper;

public ProductDao(Context context) {
    dbHelper = new ElotDbHelper(context);
}

public long insert(Product p) {
    SQLiteDatabase db = dbHelper.getWritableDatabase();
    ContentValues values = new ContentValues();
    values.put("name", p.getName());
    values.put("unit", p.getUnit());
    values.put("stock", p.getStock());
    values.put("min_stock", p.getMinStock());
    values.put("price", p.getPrice());
    return db.insert("products", null, values);
}

public Cursor getAll() {
    SQLiteDatabase db = dbHelper.getReadableDatabase();
    // SimpleCursorAdapter requiere una columna llamada '_id'
    return db.rawQuery(
        "SELECT id AS _id, name, unit, stock, min_stock, price FROM products ORDER
BY name ASC",
        null
    );
}

public int updateStock(long productId, int newStock) {
    SQLiteDatabase db = dbHelper.getWritableDatabase();
    ContentValues values = new ContentValues();
    values.put("stock", newStock);
    return db.update("products", values, "id=?", new String[]{String.valueOf(productId)});
}

public int getStockByName(String name) {
    SQLiteDatabase db = dbHelper.getReadableDatabase();
    Cursor c = db.query(
        "products",
        new String[]{"stock"},
        "name LIKE ?",
        new String[]{"%" + name + "%"},
        null, null, null
    );
    int stock = -1;
    if (c.moveToFirst()) {
        stock = c.getInt(0);
    }
    c.close();
}

```

```

        return stock;
    }

    public int getStockById(long productId) {
        SQLiteDatabase db = dbHelper.getReadableDatabase();
        Cursor c = db.query(
            "products",
            new String[]{"stock"},
            "id=?",
            new String[]{String.valueOf(productId)},
            null, null, null
        );
        int stock = 0;
        if (c.moveToFirst()) {
            stock = c.getInt(0);
        }
        c.close();
        return stock;
    }

    public Cursor getLowStock() {
        SQLiteDatabase db = dbHelper.getReadableDatabase();
        // SimpleCursorAdapter requiere una columna llamada '_id'
        return db.rawQuery(
            "SELECT id AS _id, name, unit, stock, min_stock, price FROM products " +
            "WHERE stock < min_stock ORDER BY name ASC",
            null
        );
    }

    public int update(Product p) {
        SQLiteDatabase db = dbHelper.getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put("name", p.getName());
        values.put("unit", p.getUnit());
        values.put("stock", p.getStock());
        values.put("min_stock", p.getMinStock());
        values.put("price", p.getPrice());
        return db.update("products", values, "id=?", new String[]{String.valueOf(p.getId())});
    }

    public int delete(long productId) {
        SQLiteDatabase db = dbHelper.getWritableDatabase();
        return db.delete("products", "id=?", new String[]{String.valueOf(productId)});
    }

    public Product getById(long productId) {
        SQLiteDatabase db = dbHelper.getReadableDatabase();

```

```

        Cursor c = db.query(
            "products",
            null,
            "id=?",
            new String[]{String.valueOf(productId)},
            null, null, null
        );
        Product product = null;
        if (c.moveToFirst()) {
            product = new Product(
                c.getLong(c.getColumnIndexOrThrow("id")),
                c.getString(c.getColumnIndexOrThrow("name")),
                c.getString(c.getColumnIndexOrThrow("unit")),
                c.getInt(c.getColumnIndexOrThrow("stock")),
                c.getInt(c.getColumnIndexOrThrow("min_stock")),
                c.getDouble(c.getColumnIndexOrThrow("price"))
            );
        }
        c.close();
        return product;
    }

    public Cursor searchByName(String query) {
        SQLiteDatabase db = dbHelper.getReadableDatabase();
        // SimpleCursorAdapter requiere una columna llamada '_id'
        return db.rawQuery(
            "SELECT id AS _id, name, unit, stock, min_stock, price FROM products " +
            "WHERE name LIKE ? OR unit LIKE ? ORDER BY name ASC",
            new String[]{"%" + query + "%", "%" + query + "%"}
        );
    }
}

```

..

Archivo: java\com\example\proyectofinalappmobile\data\db\DatabaseInitializer.java

``java

```
package com.example.proyectofinalappmobile.data.db;
```

```
import android.content.Context;
```

```
import com.example.proyectofinalappmobile.data.dao.ProductDao;
```

```
import com.example.proyectofinalappmobile.data.models.Product;
```

```

/**
 * Helper para inicializar la base de datos con datos de prueba.
 * Ãstil para desarrollo y testing.
 */
public class DatabaseInitializer {

    public static void populateSampleData(Context context) {
        ProductDao productDao = new ProductDao(context);

        // Verificar si ya hay datos
        if (productDao.getAll().getCount() > 0) {
            return; // Ya hay datos, no inicializar
        }

        // Productos tÃpicos de un puesto de elotes
        Product[] sampleProducts = {
            new Product(0, "Elotes", "piezas", 50, 20, 15.0),
            new Product(0, "Esquites", "vasos", 30, 15, 20.0),
            new Product(0, "Mayonesa", "frascos", 5, 3, 45.0),
            new Product(0, "Queso rallado", "bolsas", 8, 5, 35.0),
            new Product(0, "Chile en polvo", "bolsas", 4, 2, 25.0),
            new Product(0, "Limonas", "kg", 2, 1, 30.0),
            new Product(0, "Mantequilla", "barras", 6, 3, 22.0),
            new Product(0, "Sal", "bolsas", 3, 1, 15.0),
            new Product(0, "Vasos desechables", "paquetes", 10, 5, 40.0),
            new Product(0, "Servilletas", "paquetes", 8, 4, 18.0)
        };

        for (Product product : sampleProducts) {
            productDao.insert(product);
        }
    }

    public static void clearAllData(Context context) {
        // Eliminar la base de datos completa
        context.deleteDatabase("elotapp.db");
    }
}

..

## Archivo: java\com\example\proyectofinalappmobile\data\db\ElotDbHelper.java

``java

```



```

package com.example.proyectofinalappmobile.data.db;

import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class ElotDbHelper extends SQLiteOpenHelper {

    private static final String DB_NAME = "elotapp.db";
    private static final int DB_VERSION = 1;

    public ElotDbHelper(Context context) {
        super(context, DB_NAME, null, DB_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL(
            "CREATE TABLE products (" +
            "id INTEGER PRIMARY KEY AUTOINCREMENT," +
            "name TEXT NOT NULL," +
            "unit TEXT," +
            "stock INTEGER NOT NULL DEFAULT 0," +
            "min_stock INTEGER NOT NULL DEFAULT 0," +
            "price REAL NOT NULL DEFAULT 0" +
            ");"
        );

        db.execSQL(
            "CREATE TABLE movements (" +
            "id INTEGER PRIMARY KEY AUTOINCREMENT," +
            "product_id INTEGER NOT NULL," +
            "type TEXT NOT NULL," +
            "qty INTEGER NOT NULL," +
            "date INTEGER NOT NULL," +
            "FOREIGN KEY(product_id) REFERENCES products(id)" +
            ");"
        );
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        // futuro: ALTER TABLE si cambias el esquema
    }
}

```

``

Archivo: java\com\example\proyectofinalappmobile\data\models\ChatIntent.java

``java

```
package com.example.proyectofinalappmobile.data.models;
```

```
public class ChatIntent {
    public enum IntentType {
        GET_STOCK,
        GET_WEEKLY_SALES,
        GET_LOW_STOCK,
        DIRECT_ANSWER,
        UNKNOWN
    }

    private IntentType type;
    private String productName;
    private long productId;
    private String directAnswer;

    public ChatIntent() {
        this.type = IntentType.UNKNOWN;
    }

    public ChatIntent(IntentType type) {
        this.type = type;
    }

    public ChatIntent(IntentType type, String productName) {
        this.type = type;
        this.productName = productName;
    }

    public ChatIntent(IntentType type, String productName, long productId) {
        this.type = type;
        this.productName = productName;
        this.productId = productId;
    }

    public IntentType getType() {
        return type;
    }

    public void setType(IntentType type) {
```

```

        this.type = type;
    }

    public String getProductName() {
        return productName;
    }

    public void setProductName(String productName) {
        this.productName = productName;
    }

    public long getProductId() {
        return productId;
    }

    public void setProductId(long productId) {
        this.productId = productId;
    }

    public String getDirectAnswer() {
        return directAnswer;
    }

    public void setDirectAnswer(String directAnswer) {
        this.directAnswer = directAnswer;
    }
}

..

```

Archivo: java\com\example\proyectofinalappmobile\data\models\Movement.java

```
``java
```

```

package com.example.proyectofinalappmobile.data.models;

public class Movement {
    private long id;
    private long productId;
    private String type; // IN, OUT, SALE
    private int qty;
    private long date; // timestamp

    public Movement() {
    }
}

```

```
public Movement(long productId, String type, int qty, long date) {  
    this.productId = productId;  
    this.type = type;  
    this.qty = qty;  
    this.date = date;  
}
```

```
public Movement(long id, long productId, String type, int qty, long date) {  
    this.id = id;  
    this.productId = productId;  
    this.type = type;  
    this.qty = qty;  
    this.date = date;  
}
```

```
public long getId() {  
    return id;  
}
```

```
public void setId(long id) {  
    this.id = id;  
}
```

```
public long getProductId() {  
    return productId;  
}
```

```
public void setProductId(long productId) {  
    this.productId = productId;  
}
```

```
public String getType() {  
    return type;  
}
```

```
public void setType(String type) {  
    this.type = type;  
}
```

```
public int getQty() {  
    return qty;  
}
```

```
public void setQty(int qty) {  
    this.qty = qty;  
}
```

```

    public long getDate() {
        return date;
    }

    public void setDate(long date) {
        this.date = date;
    }
}

..

```

Archivo: java\com\example\proyectofinalappmobile\data\models\Product.java

```

`java

```

```

package com.example.proyectofinalappmobile.data.models;

public class Product {
    private long id;
    private String name;
    private String unit;
    private int stock;
    private int minStock;
    private double price;

    public Product() {
    }

    public Product(long id, String name, String unit, int stock, int minStock, double price) {
        this.id = id;
        this.name = name;
        this.unit = unit;
        this.stock = stock;
        this.minStock = minStock;
        this.price = price;
    }

    public long getId() {
        return id;
    }

    public void setId(long id) {
        this.id = id;
    }
}

```

```

public String getName() {
    return name;
}

public void setName(String name) {
    this.name = name;
}

public String getUnit() {
    return unit;
}

public void setUnit(String unit) {
    this.unit = unit;
}

public int getStock() {
    return stock;
}

public void setStock(int stock) {
    this.stock = stock;
}

public int getMinStock() {
    return minStock;
}

public void setMinStock(int minStock) {
    this.minStock = minStock;
}

public double getPrice() {
    return price;
}

public void setPrice(double price) {
    this.price = price;
}
}

..

```

Archivo: java\com\example\proyectofinalappmobile\domain\ChatController.java

```
``java
```

```
package com.example.proyectofinalappmobile.domain;
```

```
import android.content.Context;
import android.database.Cursor;
```

```
import com.example.proyectofinalappmobile.data.models.ChatIntent;
```

```
public class ChatController {
```

```
    private final InventoryController inventoryController;
    private final MovementController movementController;
```

```
    public ChatController(Context context) {
        inventoryController = new InventoryController(context);
        movementController = new MovementController(context);
    }
```

```
    public String handleIntent(ChatIntent intent) {
        switch (intent.getType()) {
            case DIRECT_ANSWER:
                // Respuesta directa de Gemini
                return intent.getDirectAnswer();

            case GET_STOCK:
                int stock = inventoryController.getStockByName(intent.getProductName());
                if (stock < 0) {
                    return "No encontrÃ© ese producto.";
                }
                return "Tienes " + stock + " " + intent.getProductName() + ".";

```

```
            case GET_WEEKLY_SALES:
                long productId = intent.getProductId();
                int total = movementController.getWeeklySales(productId);
                return "Vendiste " + total + " " + intent.getProductName() + " en la Ãºltima
semana.";

```

```
            case GET_LOW_STOCK:
                Cursor c = inventoryController.getLowStock();
                if (!c.moveToFirst()) {
                    c.close();
                    return "No hay productos debajo del mÃnimo.";
                }
                StringBuilder sb = new StringBuilder("Productos bajo mÃnimo: ");
                do {
                    String name = c.getString(c.getColumnIndexOrThrow("name"));
                    int s = c.getInt(c.getColumnIndexOrThrow("stock"));

```

```

        int min = c.getInt(c.getColumnIndexOrThrow("min_stock"));
        sb.append(name).append(" ").append(s).append("/").append(min).append(",
");
    } while (c.moveToNext());
    c.close();
    return sb.toString();

    default:
        return "No entendÃ tu pregunta.";
    }
}
}

```

``

Archivo: java\com\example\proyectofinalappmobile\domain\InventoryController.java

``java

```
package com.example.proyectofinalappmobile.domain;
```

```
import android.content.Context;
import android.database.Cursor;
```

```
import com.example.proyectofinalappmobile.data.dao.ProductDao;
```

```
public class InventoryController {
```

```
    private final ProductDao productDao;
```

```
    public InventoryController(Context context) {
        productDao = new ProductDao(context);
    }
```

```
    public Cursor getAllProducts() {
        return productDao.getAll();
    }
```

```
    public int getStockByName(String name) {
        return productDao.getStockByName(name);
    }
```

```
    public Cursor getLowStock() {
        return productDao.getLowStock();
    }
```



```

    public Cursor searchProducts(String query) {
        return productDao.searchByName(query);
    }
}

```

``

Archivo: java\com\example\proyectofinalappmobile\domain\MovementController.java

``java

```

package com.example.proyectofinalappmobile.domain;

import android.content.Context;

import com.example.proyectofinalappmobile.data.dao.MovementDao;
import com.example.proyectofinalappmobile.data.dao.ProductDao;
import com.example.proyectofinalappmobile.data.models.Movement;

public class MovementController {

    private final MovementDao movementDao;
    private final ProductDao productDao;

    public MovementController(Context context) {
        movementDao = new MovementDao(context);
        productDao = new ProductDao(context);
    }

    public void registerMovement(long productId, String type, int qty) {
        long now = System.currentTimeMillis();
        Movement m = new Movement(productId, type, qty, now);
        movementDao.insert(m);

        // actualiza stock
        int currentStock = productDao.getStockById(productId);
        int newStock = currentStock;
        if ("IN".equals(type)) {
            newStock += qty;
        } else {
            newStock -= qty;
        }
        productDao.updateStock(productId, newStock);
    }
}

```

```

    public int getWeeklySales(long productId) {
        return movementDao.getSalesForLastDays(productId, 7);
    }
}

```

``

Archivo: java\com\example\proyectofinalappmobile\services\StockCheckWorker.java

``java

```

package com.example.proyectofinalappmobile.services;

```

```

import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.app.PendingIntent;
import android.content.Context;
import android.content.Intent;
import android.database.Cursor;
import android.os.Build;

```

```

import androidx.annotation.NonNull;
import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;
import androidx.work.Worker;
import androidx.work.WorkerParameters;

```

```

import com.example.proyectofinalappmobile.MainActivity;
import com.example.proyectofinalappmobile.R;
import com.example.proyectofinalappmobile.domain.InventoryController;

```

```

/**
 * Worker que revisa periódicamente el inventario bajo stock
 * y envía notificaciones al usuario.
 */

```

```

public class StockCheckWorker extends Worker {

```

```

    private static final String CHANNEL_ID = "stock_channel";
    private static final int NOTIFICATION_ID = 1001;

```

```

    public StockCheckWorker(@NonNull Context context, @NonNull WorkerParameters
params) {
        super(context, params);
    }
}

```

```

@NonNull
@Override
public Result doWork() {
    Context context = getApplicationContext();
    InventoryController controller = new InventoryController(context);
    Cursor c = controller.getLowStock();

    if (!c.moveToFirst()) {
        c.close();
        return Result.success();
    }

    // Contar cuántos productos están bajos
    int count = c.getCount();
    StringBuilder productList = new StringBuilder();

    do {
        String name = c.getString(c.getColumnIndexOrThrow("name"));
        int stock = c.getInt(c.getColumnIndexOrThrow("stock"));
        int minStock = c.getInt(c.getColumnIndexOrThrow("min_stock"));
        productList.append("â€¢ ").append(name).append("
").append(stock).append("/").append(minStock).append("
");
    } while (c.moveToNext());
    c.close();

    // Crear canal de notificación
    createNotificationChannel(context);

    // Intent para abrir la app cuando toquen la notificación
    Intent intent = new Intent(context, MainActivity.class);
    intent.setFlags(Intent.FLAG_ACTIVITY_NEW_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TASK);
    PendingIntent pendingIntent = PendingIntent.getActivity(
        context, 0, intent, PendingIntent.FLAG_IMMUTABLE);

    // Crear la notificación
    NotificationCompat.Builder builder =
        new NotificationCompat.Builder(context, CHANNEL_ID)
            .setSmallIcon(android.R.drawable.ic_dialog_alert)
            .setContentTitle("âš¡ Â¡AtenciÃ³n! Stock bajo")
            .setContentText(count + " producto(s) necesitan reabastecimiento")
            .setStyle(new NotificationCompat.BigTextStyle()
                .bigText("Productos con poco inventario:\n\n" + productList.toString()))
            .setPriority(NotificationCompat.PRIORITY_HIGH)
            .setContentIntent(pendingIntent)
            .setAutoCancel(true);

```

```

        NotificationManagerCompat manager = NotificationManagerCompat.from(context);
        try {
            manager.notify(NOTIFICATION_ID, builder.build());
        } catch (SecurityException e) {
            // El usuario no ha dado permiso de notificaciones
            e.printStackTrace();
        }

        return Result.success();
    }

    private void createNotificationChannel(Context context) {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            CharSequence name = "Alertas de Stock";
            String description = "Te avisa cuando tus productos estÃ¡n bajos de inventario";
            int importance = NotificationManager.IMPORTANCE_HIGH;

            NotificationChannel channel = new NotificationChannel(CHANNEL_ID, name,
            importance);
            channel.setDescription(description);

            NotificationManager notificationManager =
            context.getSystemService(NotificationManager.class);
            notificationManager.createNotificationChannel(channel);
        }
    }
}

..

```

Archivo: java\com\example\proyectofinalappmobile\services\StockService.java

```
``java
```

```
package com.example.proyectofinalappmobile.services;
```

```

import android.app.IntentService;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.content.Intent;
import android.database.Cursor;
import android.os.Build;

```

```

import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;

```

```
import com.example.proyectofinalappmobile.R;
import com.example.proyectofinalappmobile.domain.InventoryController;
```

```
/**
```

```
 * Servicio que verifica periódicamente el inventario bajo stock
```

```
 * y envía notificaciones al usuario.
```

```
 */
```

```
public class StockService extends IntentService {
```

```
    private static final String CHANNEL_ID = "stock_channel";
```

```
    private static final int NOTIFICATION_ID = 1001;
```

```
    public StockService() {
```

```
        super("StockService");
```

```
    }
```

```
    @Override
```

```
    protected void onHandleIntent(Intent intent) {
```

```
        InventoryController controller = new InventoryController(this);
```

```
        Cursor c = controller.getLowStock();
```

```
        if (!c.moveToFirst()) {
```

```
            c.close();
```

```
            return;
```

```
        }
```

```
        // Contar cuántos productos están bajos
```

```
        int count = c.getCount();
```

```
        StringBuilder productList = new StringBuilder();
```

```
        do {
```

```
            String name = c.getString(c.getColumnIndexOrThrow("name"));
```

```
            int stock = c.getInt(c.getColumnIndexOrThrow("stock"));
```

```
            int minStock = c.getInt(c.getColumnIndexOrThrow("min_stock"));
```

```
            productList.append(name).append("
```

```
(").append(stock).append("/").append(minStock).append("), ");
```

```
        } while (c.moveToNext());
```

```
        c.close();
```

```
        // Crear canal de notificación (requerido en Android 8.0+)
```

```
        createNotificationChannel();
```

```
        // Crear la notificación
```

```
        NotificationCompat.Builder builder =
```

```
            new NotificationCompat.Builder(this, CHANNEL_ID)
```

```
                .setSmallIcon(android.R.drawable.ic_dialog_alert)
```

```
                .setContentTitle("¡ Inventario bajo")
```

```

        .setContentView(count + " producto(s) por debajo del mÃnimo")
        .setStyle(new NotificationCompat.BigTextStyle()
            .bigText(productList.toString()))
        .setPriority(NotificationCompat.PRIORITY_DEFAULT)
        .setAutoCancel(true);

    NotificationManagerCompat manager = NotificationManagerCompat.from(this);
    manager.notify(NOTIFICATION_ID, builder.build());
}

private void createNotificationChannel() {
    // Solo necesario en Android 8.0 (API 26) y superior
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        CharSequence name = "Stock Alerts";
        String description = "Notificaciones de inventario bajo";
        int importance = NotificationManager.IMPORTANCE_DEFAULT;

        NotificationChannel channel = new NotificationChannel(CHANNEL_ID, name,
            importance);
        channel.setDescription(description);

        NotificationManager notificationManager =
            getSystemService(NotificationManager.class);
        notificationManager.createNotificationChannel(channel);
    }
}
}
}

```

Archivo: java\com\example\proyectofinalappmobile\ui\chat\ChatIAActivity.java

```
``java
```

```
package com.example.proyectofinalappmobile.ui.chat;
```

```
import android.os.Bundle;
import android.widget.EditText;
import android.widget.ScrollView;
import android.widget.TextView;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import com.example.proyectofinalappmobile.R;
import com.example.proyectofinalappmobile.ai.AiCallback;
```

```
import com.example.proyectofinalappmobile.ai.AiClient;
import com.example.proyectofinalappmobile.ai.GeminiAiClient;
import com.example.proyectofinalappmobile.data.models.ChatIntent;
import com.example.proyectofinalappmobile.domain.ChatController;
```

```
public class ChatIAActivity extends AppCompatActivity {
```

```
    private TextView txtHistory;
    private EditText edtQuestion;
    private ScrollView scrollView;
    private ChatController chatController;
    private AiClient aiClient;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_chat_ia);
```

```
        txtHistory = findViewById(R.id.txt_chat_history);
        edtQuestion = findViewById(R.id.edt_question);
        scrollView = findViewById(R.id.scroll_chat);
```

```
        chatController = new ChatController(this);
        aiClient = new GeminiAiClient(this);
```

```
        // Mensaje de bienvenida más amigable
        appendBot("¡Hola! Estoy aquí para ayudarte con tu negocio.\n\n" +
            "Puedes preguntarme cosas como:\n" +
            "¿Cuántos elotes tengo?\n" +
            "¿Cuánto vendrá esta semana?\n" +
            "¿Qué productos están bajos?\n\n" +
            "Pregunta lo que quieras! ¡!");
```

```
        findViewById(R.id.btn_send).setOnClickListener(v -> {
            String question = edtQuestion.getText().toString().trim();
            if (question.isEmpty()) return;
            appendUser(question);
            askla(question);
            edtQuestion.setText("");
        });
    }
```

```
    private void appendUser(String text) {
        txtHistory.append("\n\nTú: " + text);
        scrollToBottom();
    }
```

```
    private void appendBot(String text) {
```

```

        txtHistory.append("\n\nðŸ– Asistente:\n" + text);
        scrollToBottom();
    }

    private void scrollToBottom() {
        scrollView.post(() -> scrollView.fullScroll(ScrollView.FOCUS_DOWN));
    }

    private void askla(String question) {
        aiClient.ask(question, new AiCallback() {
            @Override
            public void onResult(ChatIntent intent) {
                String answer = chatController.handleIntent(intent);
                runOnUiThread(() -> appendBot(answer));
            }

            @Override
            public void onError(Exception e) {
                runOnUiThread(() -> appendBot("ðŸ• Lo siento, tuve un problema. ¿Puedes
intentar de nuevo?"));
            }
        });
    }
}

..

```

Archivo: java\com\example\proyectofinalappmobile\ui\main\ProductAdapter.java

```

`java

```

```

package com.example.proyectofinalappmobile.ui.main;

import android.content.Context;
import android.database.Cursor;
import android.graphics.Color;
import android.graphics.drawable.GradientDrawable;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.CursorAdapter;
import android.widget.TextView;

import com.example.proyectofinalappmobile.R;

```



```

public class ProductAdapter extends CursorAdapter {

    public ProductAdapter(Context context, Cursor cursor) {
        super(context, cursor, 0);
    }

    @Override
    public View newView(Context context, Cursor cursor, ViewGroup parent) {
        return LayoutInflater.from(context).inflate(R.layout.item_product, parent, false);
    }

    @Override
    public void bindView(View view, Context context, Cursor cursor) {
        TextView txtName = view.findViewById(R.id.txt_name);
        TextView txtStock = view.findViewById(R.id.txt_stock);
        TextView txtPrice = view.findViewById(R.id.txt_price);
        View indicator = view.findViewById(R.id.indicator_color);

        String name = cursor.getString(cursor.getColumnIndexOrThrow("name"));
        int stock = cursor.getInt(cursor.getColumnIndexOrThrow("stock"));
        int minStock = cursor.getInt(cursor.getColumnIndexOrThrow("min_stock"));
        double price = cursor.getDouble(cursor.getColumnIndexOrThrow("price"));
        String unit = cursor.getString(cursor.getColumnIndexOrThrow("unit"));

        // Mostrar nombre
        txtName.setText(name);

        // Mostrar stock de forma amigable
        String unitText = (unit != null && !unit.isEmpty()) ? unit : "unidades";
        txtStock.setText("ðŸ“¡ Tengo: " + stock + " " + unitText + " | ðŸ“” MÃ¡ximo: " + minStock);

        // Mostrar precio
        txtPrice.setText("$" + String.format("%.2f", price));

        // Color del indicador segÃºn nivel de stock
        int color;
        if (stock == 0) {
            color = Color.rgb(244, 67, 54); // Rojo - sin stock
        } else if (stock < minStock) {
            color = Color.rgb(255, 152, 0); // Naranja - bajo stock
        } else if (stock < minStock * 1.5) {
            color = Color.rgb(255, 235, 59); // Amarillo - advertencia
        } else {
            color = Color.rgb(76, 175, 80); // Verde - OK
        }

        // Aplicar color al cÃrculo indicador
        GradientDrawable drawable = (GradientDrawable) indicator.getBackground();
    }
}

```

```
        drawable.setColor(color);
    }
}
```

```
``
```

Archivo:

java\com\example\proyectofinalappmobile\ui\movements\MovementsActivity.java

```
``java
```

```
package com.example.proyectofinalappmobile.ui.movements;
```

```
import android.database.Cursor;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.Spinner;
import android.widget.Toast;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import com.example.proyectofinalappmobile.R;
import com.example.proyectofinalappmobile.data.dao.ProductDao;
import com.example.proyectofinalappmobile.domain.MovementController;
import com.google.android.material.textfield.TextInputEditText;
```

```
import java.util.ArrayList;
import java.util.List;
```

```
public class MovementsActivity extends AppCompatActivity {
```

```
    private Spinner spinnerType, spinnerProduct;
    private TextInputEditText edtQuantity;
    private Button btnRegister;
    private MovementController movementController;
    private ProductDao productDao;
    private List<Long> productIds;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_movements);
```

```
        movementController = new MovementController(this);
```

```

productDao = new ProductDao(this);

spinnerType = findViewById(R.id.spinner_type);
spinnerProduct = findViewById(R.id.spinner_product);
edtQuantity = findViewById(R.id.edt_quantity);
btnRegister = findViewById(R.id.btn_register);

setupTypeSpinner();
setupProductSpinner();

btnRegister.setOnClickListener(v -> registerMovement());
}

private void setupTypeSpinner() {
    // Usar strings del resources para textos amigables
    String[] types = getResources().getStringArray(R.array.movement_types);
    ArrayAdapter<String> adapter = new ArrayAdapter<>(
        this,
        android.R.layout.simple_spinner_item,
        types
    );
    adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
    spinnerType.setAdapter(adapter);
}

private void setupProductSpinner() {
    Cursor cursor = productDao.getAll();
    List<String> productNames = new ArrayList<>();
    productIds = new ArrayList<>();

    if (cursor.moveToFirst()) {
        do {
            long id = cursor.getLong(cursor.getColumnIndexOrThrow("_id"));
            String name = cursor.getString(cursor.getColumnIndexOrThrow("name"));
            productIds.add(id);
            productNames.add(name);
        } while (cursor.moveToNext());
    }
    cursor.close();

    ArrayAdapter<String> adapter = new ArrayAdapter<>(
        this,
        android.R.layout.simple_spinner_item,
        productNames
    );
    adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
    spinnerProduct.setAdapter(adapter);
}

```

```

private void registerMovement() {
    if (productIds.isEmpty()) {
        Toast.makeText(this, "No hay productos. Agrega uno primero.",
Toast.LENGTH_SHORT).show();
        return;
    }

    String quantityStr = edtQuantity.getText().toString().trim();
    if (quantityStr.isEmpty()) {
        Toast.makeText(this, "Ingresa la cantidad", Toast.LENGTH_SHORT).show();
        return;
    }

    int selectedTypePos = spinnerType.getSelectedItemPosition();
    String type;
    switch (selectedTypePos) {
        case 0:
            type = "IN";
            break;
        case 1:
            type = "OUT";
            break;
        case 2:
            type = "SALE";
            break;
        default:
            type = "IN";
    }

    int selectedProductPos = spinnerProduct.getSelectedItemPosition();
    long productId = productIds.get(selectedProductPos);
    int quantity = Integer.parseInt(quantityStr);

    movementController.registerMovement(productId, type, quantity);

    Toast.makeText(this, "Movimiento registrado", Toast.LENGTH_SHORT).show();
    finish();
}
}

..

```

Archivo: java\com\example\proyectofinalappmobile\ui\product\ProductDetailActivity.java

```
``java
```

```
package com.example.proyectofinalappmobile.ui.product;
```

```
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import com.example.proyectofinalappmobile.R;
import com.example.proyectofinalappmobile.data.dao.ProductDao;
import com.example.proyectofinalappmobile.data.models.Product;
import com.google.android.material.textfield.TextInputEditText;
```

```
public class ProductDetailActivity extends AppCompatActivity {
```

```
    private TextInputEditText edtName, edtUnit, edtStock, edtMinStock, edtPrice;
    private Button btnSave, btnDelete;
    private ProductDao productDao;
    private long productId = -1;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_product_detail);
```

```
        productDao = new ProductDao(this);
```

```
        edtName = findViewById(R.id.edt_name);
        edtUnit = findViewById(R.id.edt_unit);
        edtStock = findViewById(R.id.edt_stock);
        edtMinStock = findViewById(R.id.edt_min_stock);
        edtPrice = findViewById(R.id.edt_price);
        btnSave = findViewById(R.id.btn_save);
        btnDelete = findViewById(R.id.btn_delete);
```

```
        // Verificar si viene un producto para editar
        if (getIntent().hasExtra("product_id")) {
            productId = getIntent().getLongExtra("product_id", -1);
            loadProduct(productId);
        } else {
            btnDelete.setEnabled(false);
        }
    }
```

```
    btnSave.setOnClickListener(v -> saveProduct());
    btnDelete.setOnClickListener(v -> deleteProduct());
```

```
}
```

```

private void loadProduct(long id) {
    Product product = productDao.getByld(id);
    if (product != null) {
        edtName.setText(product.getName());
        edtUnit.setText(product.getUnit());
        edtStock.setText(String.valueOf(product.getStock()));
        edtMinStock.setText(String.valueOf(product.getMinStock()));
        edtPrice.setText(String.valueOf(product.getPrice()));
    }
}

private void saveProduct() {
    String name = edtName.getText().toString().trim();
    String unit = edtUnit.getText().toString().trim();
    String stockStr = edtStock.getText().toString().trim();
    String minStockStr = edtMinStock.getText().toString().trim();
    String priceStr = edtPrice.getText().toString().trim();

    if (name.isEmpty()) {
        Toast.makeText(this, "El nombre es obligatorio", Toast.LENGTH_SHORT).show();
        return;
    }

    int stock = stockStr.isEmpty() ? 0 : Integer.parseInt(stockStr);
    int minStock = minStockStr.isEmpty() ? 0 : Integer.parseInt(minStockStr);
    double price = priceStr.isEmpty() ? 0.0 : Double.parseDouble(priceStr);

    Product product = new Product(productId, name, unit, stock, minStock, price);

    if (productId == -1) {
        // Nuevo producto
        long result = productDao.insert(product);
        if (result > 0) {
            Toast.makeText(this, "Producto guardado", Toast.LENGTH_SHORT).show();
            finish();
        }
    } else {
        // Actualizar producto
        int result = productDao.update(product);
        if (result > 0) {
            Toast.makeText(this, "Producto actualizado", Toast.LENGTH_SHORT).show();
            finish();
        }
    }
}

private void deleteProduct() {

```

```

        if (productId != -1) {
            int result = productDao.delete(productId);
            if (result > 0) {
                Toast.makeText(this, "Producto eliminado", Toast.LENGTH_SHORT).show();
                finish();
            }
        }
    }
}

```

``

Archivo: java\com\example\proyectofinalappmobile\HistoryActivity.java

``java

```
package com.example.proyectofinalappmobile;
```

```

import android.app.AlertDialog;
import android.content.SharedPreferences;
import android.database.Cursor;
import android.os.Bundle;
import android.widget.Button;
import android.widget.ListView;
import android.widget.SimpleCursorAdapter;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import com.example.proyectofinalappmobile.data.dao.MovementDao;
import com.example.proyectofinalappmobile.data.dao.ProductDao;
import com.example.proyectofinalappmobile.data.models.Movement;
import com.example.proyectofinalappmobile.domain.MovementController;
import java.text.NumberFormat;
import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.Locale;

```

```

public class HistoryActivity extends AppCompatActivity {
    private MovementController movementController;
    private MovementDao movementDao;
    private ListView listMovements;
    private TextView txtWeeklySales, txtWeeklyRevenue, txtTotalMovements, txtPeriodLabel,
txtHistoryLabel;
    private Button btnFilterWeek, btnFilterMonth, btnFilterAll, btnResetWeek;
    private SharedPreferences preferences;

```

```

private String currentFilter = "week"; // week, month, all
private static final String PREFS_NAME = "ElotAppPrefs";
private static final String KEY_WEEK_START = "week_start_date";

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_history);

    movementController = new MovementController(this);
    movementDao = new MovementDao(this);
    preferences = getSharedPreferences(PREFS_NAME, MODE_PRIVATE);

    // Si es la primera vez, guardar la fecha actual como inicio de semana
    if (!preferences.contains(KEY_WEEK_START)) {
        preferences.edit().putLong(KEY_WEEK_START, System.currentTimeMillis()).apply();
    }

    listMovements = findViewById(R.id.list_movements);
    txtWeeklySales = findViewById(R.id.txt_weekly_sales);
    txtWeeklyRevenue = findViewById(R.id.txt_weekly_revenue);
    txtTotalMovements = findViewById(R.id.txt_total_movements);
    txtPeriodLabel = findViewById(R.id.txt_period_label);
    txtHistoryLabel = findViewById(R.id.txt_history_label);

    btnFilterWeek = findViewById(R.id.btn_filter_week);
    btnFilterMonth = findViewById(R.id.btn_filter_month);
    btnFilterAll = findViewById(R.id.btn_filter_all);
    btnResetWeek = findViewById(R.id.btn_reset_week);

    setupFilterButtons();
    updateFilterButtonStyles(); // Aplicar estilos iniciales
    loadStatistics();
    loadMovements();
    setupMovementActions();
}

private void setupMovementActions() {
    listMovements.setOnItemLongClickListener((parent, view, position, id) -> {
        showMovementOptionsDialog(id);
        return true;
    });
}

private void showMovementOptionsDialog(long movementId) {
    new AlertDialog.Builder(this)
        .setTitle("¿Qué quieres hacer?")

```



```

    }

    // Calcular diferencia para ajustar el inventario
    int oldQty = movement.getQty();
    int difference = newQty - oldQty;

    // Actualizar el movimiento
    movement.setQty(newQty);
    dao.update(movement);

    // Ajustar el stock del producto
    ProductDao productDao = new ProductDao(this);
    int currentStock = productDao.getStockById(movement.getProductId());
    int newStock;

    if (movement.getType().equals("SALE") || movement.getType().equals("OUT")) {
        newStock = currentStock - difference; // Si vendo mÃ¡s, resto mÃ¡s
    } else {
        newStock = currentStock + difference; // Si entra mÃ¡s, sumo mÃ¡s
    }

    productDao.updateStock(movement.getProductId(), newStock);

    Toast.makeText(this, "âœ… Cantidad actualizada",
Toast.LENGTH_SHORT).show();
    loadStatistics();
    loadMovements();

    } catch (NumberFormatException e) {
        Toast.makeText(this, "âœ… Cantidad invÃ¡lida",
Toast.LENGTH_SHORT).show();
    }
})
.setNegativeButton("Cancelar", null)
.show();
}

private void deleteMovement(long movementId) {
    MovementDao dao = new MovementDao(this);
    Movement movement = dao.getById(movementId);

    if (movement == null) {
        Toast.makeText(this, "âœ… No se encontrÃ³ el movimiento",
Toast.LENGTH_SHORT).show();
        return;
    }

    new AlertDialog.Builder(this)

```

```

.setTitle("Eliminar Movimiento")
.setMessage("¿Estás seguro de eliminar este movimiento?\n\n" +
    "Cantidad: " + movement.getQty() + "\n" +
    "Tipo: " + getMovementTypeName(movement.getType()) + "\n\n" +
    "Esto también ajustará tu inventario.")
.setPositiveButton("Sí, eliminar", (dialog, which) -> {
    // Ajustar el stock antes de eliminar
    ProductDao productDao = new ProductDao(this);
    int currentStock = productDao.getStockById(movement.getProductId());
    int newStock;

    if (movement.getType().equals("SALE") || movement.getType().equals("OUT")) {
        newStock = currentStock + movement.getQty(); // Devolver al inventario
    } else {
        newStock = currentStock - movement.getQty(); // Quitar del inventario
    }

    productDao.updateStock(movement.getProductId(), newStock);

    // Eliminar el movimiento
    dao.delete(movementId);

    Toast.makeText(this, "Movimiento eliminado",
        Toast.LENGTH_SHORT).show();
    loadStatistics();
    loadMovements();
})
.setNegativeButton("Cancelar", null)
.show();
}

private String getMovementTypeName(String type) {
    switch (type) {
        case "SALE": return "Venta";
        case "IN": return "Entrada";
        case "OUT": return "Salida";
        default: return type;
    }
}

private void setupFilterButtons() {
    btnFilterWeek.setOnClickListener(v -> {
        currentFilter = "week";
        updateFilterButtonStyles();
        loadStatistics();
        loadMovements();
    });
}

```

```

        btnFilterMonth.setOnClickListener(v -> {
            currentFilter = "month";
            updateFilterButtonStyles();
            loadStatistics();
            loadMovements();
        });

        btnFilterAll.setOnClickListener(v -> {
            currentFilter = "all";
            updateFilterButtonStyles();
            loadStatistics();
            loadMovements();
        });

        btnResetWeek.setOnClickListener(v -> showResetDialog());
    }

    private void updateFilterButtonStyles() {
        // Reset all to outlined style (deseleccionados)
        btnFilterWeek.setBackgroundTintList(null);
        btnFilterMonth.setBackgroundTintList(null);
        btnFilterAll.setBackgroundTintList(null);

        btnFilterWeek.setTextColor(getResources().getColor(android.R.color.darker_gray,
null));
        btnFilterMonth.setTextColor(getResources().getColor(android.R.color.darker_gray,
null));
        btnFilterAll.setTextColor(getResources().getColor(android.R.color.darker_gray, null));

        // Highlight selected (resaltar el seleccionado)
        if (currentFilter.equals("week")) {

            btnFilterWeek.setBackgroundTintList(getResources().getColorStateList(com.google.android.
material.R.color.material_dynamic_primary50, null));
            btnFilterWeek.setTextColor(getResources().getColor(android.R.color.white, null));
        } else if (currentFilter.equals("month")) {

            btnFilterMonth.setBackgroundTintList(getResources().getColorStateList(com.google.android.
material.R.color.material_dynamic_primary50, null));
            btnFilterMonth.setTextColor(getResources().getColor(android.R.color.white, null));
        } else {

            btnFilterAll.setBackgroundTintList(getResources().getColorStateList(com.google.android.mat
erial.R.color.material_dynamic_primary50, null));
            btnFilterAll.setTextColor(getResources().getColor(android.R.color.white, null));
        }
    }
}

```

```

private void showResetDialog() {
    long weekStart = preferences.getLong(KEY_WEEK_START,
System.currentTimeMillis());
    SimpleDateFormat dateFormat = new SimpleDateFormat("dd/MM/yyyy HH:mm", new
Locale("es", "MX"));
    String startDateStr = dateFormat.format(new Date(weekStart));

    new AlertDialog.Builder(this)
        .setTitle("ðŸ”” Empezar Nueva Semana")
        .setMessage("Semana actual iniciÃ³: " + startDateStr + "\n\n" +
            "¿Quieres empezar una nueva semana AHORA?\n\n" +
            "Las ventas anteriores se guardarÃ¡n en el historial pero ya no contarÃ¡n en
'Esta Semana'.")
        .setPositiveButton("SÃ, empezar ahora", (dialog, which) -> {
            // Guardar la fecha actual como nueva fecha de inicio
            long now = System.currentTimeMillis();
            preferences.edit().putLong(KEY_WEEK_START, now).apply();

            String newStartDate = dateFormat.format(new Date(now));
            Toast.makeText(this, "â€¦ Nueva semana iniciada: " + newStartDate,
Toast.LENGTH_LONG).show();

            // Recargar estadÃsticas
            loadStatistics();
            loadMovements();
        })
        .setNegativeButton("Cancelar", null)
        .show();
}

private void loadStatistics() {
    // Obtener estadÃsticas segÃºn el filtro
    MovementDao dao = new MovementDao(this);

    int salesCount;
    double revenue;

    if (currentFilter.equals("week")) {
        long weekStart = preferences.getLong(KEY_WEEK_START,
System.currentTimeMillis());
        SimpleDateFormat dateFormat = new SimpleDateFormat("dd/MM", new Locale("es",
"MX"));
        String startDateStr = dateFormat.format(new Date(weekStart));

        txtPeriodLabel.setText("Esta semana (desde " + startDateStr + "):");
        txtHistoryLabel.setText("Historial de esta semana:");
        salesCount = dao.getCustomWeeklySalesCount(weekStart);
        revenue = dao.getCustomWeeklyRevenue(weekStart);
    }
}

```

```

    } else if (currentFilter.equals("month")) {
        txtPeriodLabel.setText("Este mes:");
        txtHistoryLabel.setText("Historial de este mes:");
        salesCount = dao.getMonthlySalesCount();
        revenue = dao.getMonthlyRevenue();
    } else {
        txtPeriodLabel.setText("Desde siempre:");
        txtHistoryLabel.setText("Historial completo:");
        salesCount = dao.getTotalSalesCount();
        revenue = dao.getTotalRevenue();
    }

    int totalMovements = dao.getTotalMovementsCount();

    NumberFormat currencyFormat = NumberFormat.getCurrencyInstance(new
    Locale("es", "MX"));

    txtWeeklySales.setText("ðŸ“¡ " + salesCount + " ventas");
    txtWeeklyRevenue.setText("ðŸ’° " + currencyFormat.format(revenue));
    txtTotalMovements.setText("ðŸ“ˆ " + totalMovements + " movimientos totales");
}

private void loadMovements() {
    MovementDao dao = new MovementDao(this);
    Cursor cursor;

    if (currentFilter.equals("week")) {
        long weekStart = preferences.getLong(KEY_WEEK_START,
    System.currentTimeMillis());
        cursor = dao.getCustomWeeklyMovementsWithProductNames(weekStart);
    } else if (currentFilter.equals("month")) {
        cursor = dao.getMonthlyMovementsWithProductNames();
    } else {
        cursor = dao.getAllWithProductNames();
    }

    String[] from = {"type_icon", "product_name", "quantity_text", "date_text"};
    int[] to = {R.id.txt_movement_type, R.id.txt_movement_product,
        R.id.txt_movement_quantity, R.id.txt_movement_date};

    SimpleCursorAdapter adapter = new SimpleCursorAdapter(
        this, R.layout.item_movement, cursor, from, to, 0
    );

    listMovements.setAdapter(adapter);
}

@Override

```

```

        protected void onResume() {
            super.onResume();
            loadStatistics();
            loadMovements();
        }
    }
}

```

``

Archivo: java\com\example\proyectofinalappmobile\MainActivity.java

``java

```

package com.example.proyectofinalappmobile;

```

```

import android.content.Intent;
import android.database.Cursor;
import android.os.Bundle;
import android.text.Editable;
import android.text.TextWatcher;
import android.view.View;
import android.widget.EditText;
import android.widget.ImageButton;
import android.widget.ListView;

```

```

import androidx.appcompat.app.AppCompatActivity;
import androidx.work.ExistingPeriodicWorkPolicy;
import androidx.work.PeriodicWorkRequest;
import androidx.work.WorkManager;

```

```

import java.util.concurrent.TimeUnit;

```

```

import com.example.proyectofinalappmobile.data.db.DatabaseInitializer;
import com.example.proyectofinalappmobile.domain.InventoryController;
import com.example.proyectofinalappmobile.services.StockCheckWorker;
import com.example.proyectofinalappmobile.ui.chat.ChatIAActivity;
import com.example.proyectofinalappmobile.ui.main.ProductAdapter;
import com.example.proyectofinalappmobile.ui.product.ProductDetailActivity;
import com.google.android.material.floatingactionbutton.FloatingActionButton;

```

```

public class MainActivity extends AppCompatActivity {

```

```

    private ListView listView;
    private InventoryController inventoryController;
    private ProductAdapter adapter;

```

```

private FloatingActionButton fabMenu;
private View fabMenuLayout;
private EditText searchProducts;
private ImageButton btnClearSearch;
private boolean menuOpen = false;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    // Inicializar datos de prueba (solo la primera vez)
    DatabaseInitializer.populateSampleData(this);

    inventoryController = new InventoryController(this);

    // Configurar notificaciones automáticas de stock bajo
    setupStockNotifications();

    // Revisar stock inmediatamente
    checkStockNow();
    listView = findViewById(R.id.list_products);
    fabMenu = findViewById(R.id.fab_menu);
    fabMenuLayout = findViewById(R.id.fab_menu_layout);
    searchProducts = findViewById(R.id.search_products);
    btnClearSearch = findViewById(R.id.btn_clear_search);

    loadProducts();

    // Configurar búsqueda
    setupSearch();

    // Click en productos para editar
    listView.setOnItemClickListener((parent, view, position, id) -> {
        Intent intent = new Intent(this, ProductDetailActivity.class);
        intent.putExtra("product_id", id);
        startActivity(intent);
    });

    // Configurar menú flotante
    setupFloatingMenu();
}

private void setupSearch() {
    searchProducts.addTextChangedListener(new TextWatcher() {
        @Override
        public void beforeTextChanged(CharSequence s, int start, int count, int after) {}
    });
}

```



```

@Override
public void onTextChanged(CharSequence s, int start, int before, int count) {
    String query = s.toString().trim();

    if (query.isEmpty()) {
        btnClearSearch.setVisibility(View.GONE);
        loadProducts();
    } else {
        btnClearSearch.setVisibility(View.VISIBLE);
        searchProducts(query);
    }
}

@Override
public void afterTextChanged(Editable s) {}
});

btnClearSearch.setOnClickListener(v -> {
    searchProducts.setText("");
    btnClearSearch.setVisibility(View.GONE);
});
}

private void searchProducts(String query) {
    Cursor cursor = inventoryController.searchProducts(query);
    if (adapter == null) {
        adapter = new ProductAdapter(this, cursor);
        listView.setAdapter(adapter);
    } else {
        adapter.changeCursor(cursor);
    }
}

private void setupFloatingMenu() {
    // Bot n principal para abrir/cerrar men 
    fabMenu.setOnClickListener(v -> {
        if (menuOpen) {
            closeMenu();
        } else {
            openMenu();
        }
    });

    // Opciones del men 
    findViewById(R.id.fab_add_product).setOnClickListener(v -> {
        closeMenu();
        startActivity(new Intent(this, ProductDetailActivity.class));
    });
}

```

```

        findViewById(R.id.fab_history).setOnClickListener(v -> {
            closeMenu();
            startActivity(new Intent(this, HistoryActivity.class));
        });

        findViewById(R.id.fab_reports).setOnClickListener(v -> {
            closeMenu();
            startActivity(new Intent(this, ReportsActivity.class));
        });

        findViewById(R.id.fab_movements).setOnClickListener(v -> {
            closeMenu();
            startActivity(new Intent(this,
com.example.proyectofinalappmobile.ui.movements.MovementsActivity.class));
        });

        findViewById(R.id.fab_chat).setOnClickListener(v -> {
            closeMenu();
            startActivity(new Intent(this, ChatIAActivity.class));
        });
    }

    private void openMenu() {
        menuOpen = true;
        fabMenuLayout.setVisibility(View.VISIBLE);
        fabMenu.setImageResource(android.R.drawable.ic_menu_close_clear_cancel);
    }

    private void closeMenu() {
        menuOpen = false;
        fabMenuLayout.setVisibility(View.GONE);
        fabMenu.setImageResource(android.R.drawable.ic_menu_add);
    }

    private void loadProducts() {
        Cursor cursor = inventoryController.getAllProducts();
        if (adapter == null) {
            adapter = new ProductAdapter(this, cursor);
            listView.setAdapter(adapter);
        } else {
            adapter.changeCursor(cursor);
        }
    }

    private void setupStockNotifications() {
        // Programar verificaci3n diaria de stock bajo
        PeriodicWorkRequest stockCheckRequest = new PeriodicWorkRequest.Builder(

```

```

        StockCheckWorker.class,
        1, TimeUnit.DAYS // Revisar una vez al día
    ).build();

    WorkManager.getInstance(this).enqueueUniquePeriodicWork(
        "stock_check",
        ExistingPeriodicWorkPolicy.KEEP, // No duplicar si ya existe
        stockCheckRequest
    );
}

private void checkStockNow() {
    // Ejecutar verificación inmediata en segundo plano
    WorkManager.getInstance(this).enqueue(
        new
        androidx.work.OneTimeWorkRequest.Builder(StockCheckWorker.class).build()
    );
}

@Override
protected void onResume() {
    super.onResume();
    loadProducts();
    closeMenu(); // Cerrar menú al volver a la actividad
}

@Override
protected void onDestroy() {
    super.onDestroy();
    if (adapter != null && adapter.getCursor() != null) {
        adapter.getCursor().close();
    }
}

@Override
public void onBackPressed() {
    if (menuOpen) {
        closeMenu();
    } else {
        super.onBackPressed();
    }
}
}
..

```

Archivo: java\com\example\proyectofinalappmobile\ReportsActivity.java

```
``java
```

```
package com.example.proyectofinalappmobile;
```

```
import android.content.SharedPreferences;
import android.database.Cursor;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.Button;
import android.widget.ListView;
import android.widget.TextView;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import com.example.proyectofinalappmobile.data.dao.MovementDao;
```

```
import java.text.NumberFormat;
import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Date;
import java.util.List;
import java.util.Locale;
```

```
public class ReportsActivity extends AppCompatActivity {
```

```
    private MovementDao movementDao;
    private ListView listTopProducts;
    private ListView listDailyStats;
    private TextView txtPeriodTitle;
    private TextView txtTotalSales;
    private TextView txtTotalRevenue;
    private Button btnPeriodWeek;
    private Button btnPeriodAll;
    private boolean showingWeek = true;
    private SharedPreferences prefs;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_reports);
```

```
        movementDao = new MovementDao(this);
        prefs = getSharedPreferences("ElotAppPrefs", MODE_PRIVATE);
```

```

listTopProducts = findViewById(R.id.list_top_products);
listDailyStats = findViewById(R.id.list_daily_stats);
txtPeriodTitle = findViewById(R.id.txt_period_title);
txtTotalSales = findViewById(R.id.txt_total_sales);
txtTotalRevenue = findViewById(R.id.txt_total_revenue);
btnPeriodWeek = findViewById(R.id.btn_period_week);
btnPeriodAll = findViewById(R.id.btn_period_all);

setupButtons();
loadWeeklyData();
}

private void setupButtons() {
    btnPeriodWeek.setOnClickListener(v -> {
        showingWeek = true;
        updateButtonStyles();
        loadWeeklyData();
    });

    btnPeriodAll.setOnClickListener(v -> {
        showingWeek = false;
        updateButtonStyles();
        loadAllTimeData();
    });
}

private void updateButtonStyles() {
    // Simplemente cambiar el alpha para indicar cual está activo
    if (showingWeek) {
        btnPeriodWeek.setAlpha(1.0f);
        btnPeriodAll.setAlpha(0.5f);
    } else {
        btnPeriodWeek.setAlpha(0.5f);
        btnPeriodAll.setAlpha(1.0f);
    }
}

private void loadWeeklyData() {
    txtPeriodTitle.setText("ðŸ† Productos mÃ¡s vendidos (esta semana)");

    long weekStart = prefs.getLong("week_start_date", System.currentTimeMillis() - 7L *
24L * 60L * 60L * 1000L);

    // Productos mÃ¡s vendidos
    Cursor topProducts = movementDao.getTopSellingProductsThisWeek(10, weekStart);
    TopProductsAdapter adapter = new TopProductsAdapter(topProducts);
    listTopProducts.setAdapter(adapter);
}

```

```

// Estadísticas diarias
Cursor dailyStats = movementDao.getDailySalesStats(7);
DailyStatsAdapter statsAdapter = new DailyStatsAdapter(dailyStats);
listDailyStats.setAdapter(statsAdapter);

// Resumen
int totalSales = movementDao.getCustomWeeklySalesCount(weekStart);
double totalRevenue = movementDao.getCustomWeeklyRevenue(weekStart);

txtTotalSales.setText("ðŸ“¡ Total de ventas esta semana: " + totalSales);
txtTotalRevenue.setText("ðŸ“° Ingresos esta semana: " +
formatCurrency(totalRevenue));
}

private void loadAllTimeData() {
txtPeriodTitle.setText("ðŸ“† Productos mÃ¡s vendidos (todo el tiempo)");

// Productos mÃ¡s vendidos
Cursor topProducts = movementDao.getTopSellingProducts(10);
TopProductsAdapter adapter = new TopProductsAdapter(topProducts);
listTopProducts.setAdapter(adapter);

// Estadísticas diarias (Ãºltimos 7 dÃ­as)
Cursor dailyStats = movementDao.getDailySalesStats(7);
DailyStatsAdapter statsAdapter = new DailyStatsAdapter(dailyStats);
listDailyStats.setAdapter(statsAdapter);

// Resumen
int totalSales = movementDao.getTotalSalesCount();
double totalRevenue = movementDao.getTotalRevenue();

txtTotalSales.setText("ðŸ“¡ Total de ventas: " + totalSales);
txtTotalRevenue.setText("ðŸ“° Ingresos totales: " + formatCurrency(totalRevenue));
}

private String formatCurrency(double amount) {
    NumberFormat formatter = NumberFormat.getCurrencyInstance(new Locale("es",
"MX"));
    return formatter.format(amount);
}

// Adapter para productos mÃ¡s vendidos
class TopProductsAdapter extends BaseAdapter {
    private List<TopProductItem> items = new ArrayList<>();
    private int maxSold = 1;

    TopProductsAdapter(Cursor cursor) {

```

```

        if (cursor != null && cursor.moveToFirst()) {
            maxSold = cursor.getInt(cursor.getColumnIndexOrThrow("total_sold"));
            cursor.moveToFirst();

            int rank = 1;
            do {
                String name = cursor.getString(cursor.getColumnIndexOrThrow("name"));
                int totalSold = cursor.getInt(cursor.getColumnIndexOrThrow("total_sold"));
                double revenue =
cursor.getDouble(cursor.getColumnIndexOrThrow("total_revenue"));
                int timesSold = cursor.getInt(cursor.getColumnIndexOrThrow("times_sold"));
                String unit = cursor.getString(cursor.getColumnIndexOrThrow("unit"));

                items.add(new TopProductItem(rank++, name, totalSold, revenue, timesSold,
unit));
            } while (cursor.moveToNext());

            cursor.close();
        }
    }

    @Override
    public int getCount() {
        return items.size();
    }

    @Override
    public TopProductItem getItem(int position) {
        return items.get(position);
    }

    @Override
    public long getItemId(int position) {
        return position;
    }

    @Override
    public View getView(int position, View convertView, ViewGroup parent) {
        if (convertView == null) {
            convertView = LayoutInflater.from(ReportsActivity.this)
                .inflate(R.layout.item_top_product, parent, false);
        }

        TopProductItem item = getItem(position);

        TextView txtRank = convertView.findViewById(R.id.txt_rank);
        TextView txtProductName = convertView.findViewById(R.id.txt_product_name);
        TextView txtSoldQuantity = convertView.findViewById(R.id.txt_sold_quantity);

```

```

TextView txtRevenue = convertView.findViewById(R.id.txt_revenue);
TextView txtTimesSold = convertView.findViewById(R.id.txt_times_sold);
View progressBar = convertView.findViewById(R.id.progress_bar);

```

```

txtRank.setText(String.valueOf(item.rank));
txtProductName.setText(item.name);
txtSoldQuantity.setText("ðŸ“¡ " + item.totalSold + " " + item.unit + " vendidos");
txtRevenue.setText("ðŸ’° " + formatCurrency(item.revenue) + " en ventas");
txtTimesSold.setText("ðŸ“” „ Vendido " + item.timesSold + " veces");

```

```

// Ajustar ancho de barra de progreso
float percentage = (float) item.totalSold / maxSold;
ViewGroup.LayoutParams params = progressBar.getLayoutParams();
params.width = (int) (parent.getWidth() * percentage * 0.9);
progressBar.setLayoutParams(params);

```

```

return convertView;

```

```

}
}

```

```

// Adapter para estadísticas diarias

```

```

class DailyStatsAdapter extends BaseAdapter {
    private List<DailyStatItem> items = new ArrayList<>();
    private SimpleDateFormat inputFormat = new SimpleDateFormat("yyyy-MM-dd",
Locale.getDefault());
    private SimpleDateFormat outputFormat = new SimpleDateFormat("EEEE d 'de' MMM",
new Locale("es", "ES"));

```

```

DailyStatsAdapter(Cursor cursor) {
    if (cursor != null && cursor.moveToFirst()) {
        do {
            String day = cursor.getString(cursor.getColumnIndexOrThrow("day"));
            int salesCount = cursor.getInt(cursor.getColumnIndexOrThrow("sales_count"));
            int itemsSold = cursor.getInt(cursor.getColumnIndexOrThrow("items_sold"));
            double revenue =
cursor.getDouble(cursor.getColumnIndexOrThrow("revenue"));

```

```

            items.add(new DailyStatItem(day, salesCount, itemsSold, revenue));
        } while (cursor.moveToNext());

```

```

        cursor.close();
    }
}

```

```

@Override
public int getCount() {
    return items.size();
}

```



```

@Override
public DailyStatItem getItem(int position) {
    return items.get(position);
}

@Override
public long getItemId(int position) {
    return position;
}

@Override
public View getView(int position, View convertView, ViewGroup parent) {
    if (convertView == null) {
        convertView = LayoutInflater.from(ReportsActivity.this)
            .inflate(R.layout.item_daily_stat, parent, false);
    }

    DailyStatItem item = getItem(position);

    TextView txtDay = convertView.findViewById(R.id.txt_day);
    TextView txtSalesCount = convertView.findViewById(R.id.txt_sales_count);
    TextView txtItemsSold = convertView.findViewById(R.id.txt_items_sold);
    TextView txtDayRevenue = convertView.findViewById(R.id.txt_day_revenue);

    try {
        Date date = inputFormat.parse(item.day);
        txtDay.setText("ðŸ“ˆ... " + outputFormat.format(date));
    } catch (Exception e) {
        txtDay.setText("ðŸ“ˆ... " + item.day);
    }

    txtSalesCount.setText(item.salesCount + " ventas");
    txtItemsSold.setText("ðŸ“ˆ" + item.itemsSold + " productos");
    txtDayRevenue.setText(formatCurrency(item.revenue));

    return convertView;
}
}

// Clases de datos
static class TopProductItem {
    int rank;
    String name;
    int totalSold;
    double revenue;
    int timesSold;
    String unit;
}

```

```

        TopProductItem(int rank, String name, int totalSold, double revenue, int timesSold,
String unit) {
            this.rank = rank;
            this.name = name;
            this.totalSold = totalSold;
            this.revenue = revenue;
            this.timesSold = timesSold;
            this.unit = unit;
        }
    }
}

```

```

static class DailyStatItem {
    String day;
    int salesCount;
    int itemsSold;
    double revenue;

    DailyStatItem(String day, int salesCount, int itemsSold, double revenue) {
        this.day = day;
        this.salesCount = salesCount;
        this.itemsSold = itemsSold;
        this.revenue = revenue;
    }
}
}

```

``

Archivo: res\drawable\bg_menu_label.xml

``xml

```

<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/surface"/>
    <corners android:radius="24dp"/>
    <stroke android:width="1dp" android:color="@color/divider"/>
</shape>

```

``

Archivo: res\drawable\bg_search.xml

``xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/surface"/>
    <corners android:radius="24dp"/>
    <stroke android:width="1dp" android:color="@color/divider"/>
</shape>
```

``

Archivo: res\drawable\circle_indicator.xml

``xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="oval">
    <solid android:color="#4CAF50"/>
    <size android:width="12dp" android:height="12dp"/>
</shape>
```

``

Archivo: res\drawable\circle_rank.xml

``xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="oval">
    <solid android:color="?attr/colorPrimary"/>
</shape>
```

``

Archivo: res\drawable\header_background.xml

``xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/primary"/>
    <corners
        android:bottomLeftRadius="32dp"
        android:bottomRightRadius="32dp"/>
</shape>
```

``

Archivo: res\drawable\ic_launcher_background.xml

``xml

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="108dp"
    android:height="108dp"
    android:viewportWidth="108"
    android:viewportHeight="108">
    <path
        android:fillColor="#3DDC84"
        android:pathData="M0,0h108v108h-108z" />
    <path
        android:fillColor="#00000000"
        android:pathData="M9,0L9,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,0L19,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M29,0L29,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
```

```

        android:fillColor="#00000000"
        android:pathData="M39,0L39,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M49,0L49,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M59,0L59,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M69,0L69,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M79,0L79,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M89,0L89,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M99,0L99,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,9L108,9"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,19L108,19"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,29L108,29"
        android:strokeWidth="0.8"

```

```
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,39L108,39"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,49L108,49"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,59L108,59"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,69L108,69"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,79L108,79"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,89L108,89"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,99L108,99"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,29L89,29"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,39L89,39"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
```

```

        android:pathData="M19,49L89,49"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M19,59L89,59"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M19,69L89,69"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M19,79L89,79"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M29,19L29,89"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M39,19L39,89"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M49,19L49,89"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M59,19L59,89"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M69,19L69,89"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M79,19L79,89"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />

```

</vector>

..

Archivo: res\drawable\ic_launcher_foreground.xml

``xml

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:aapt="http://schemas.android.com/aapt"
  android:width="108dp"
  android:height="108dp"
  android:viewportWidth="108"
  android:viewportHeight="108">
  <path android:pathData="M31,63.928c0,0 6.4,-11 12.1,-13.1c7.2,-2.6 26,-1.4
26,-1.4l38.1,38.1L107,108.928l-32,-1L31,63.928z">
    <aapt:attr name="android:fillColor">
      <gradient
        android:endX="85.84757"
        android:endY="92.4963"
        android:startX="42.9492"
        android:startY="49.59793"
        android:type="linear">
        <item
          android:color="#44000000"
          android:offset="0.0" />
        <item
          android:color="#00000000"
          android:offset="1.0" />
        </gradient>
      </aapt:attr>
    </path>
    <path
      android:fillColor="#FFFFFF"
      android:fillType="nonZero"
      android:pathData="M65.3,45.828l3.8,-6.6c0.2,-0.4 0.1,-0.9 -0.3,-1.1c-0.4,-0.2 -0.9,-0.1
-1.1,0.3l-3.9,6.7c-6.3,-2.8 -13.4,-2.8 -19.7,0l-3.9,-6.7c-0.2,-0.4 -0.7,-0.5
-1.1,-0.3C38.8,38.328 38.7,38.828 38.9,39.228l3.8,6.6C36.2,49.428 31.7,56.028
31,63.928h46C76.3,56.028 71.8,49.428 65.3,45.828zM43.4,57.328c-0.8,0 -1.5,-0.5
-1.8,-1.2c-0.3,-0.7 -0.1,-1.5 0.4,-2.1c0.5,-0.5 1.4,-0.7 2.1,-0.4c0.7,0.3 1.2,1
1.2,1.8C45.3,56.528 44.5,57.328 43.4,57.328L43.4,57.328zM64.6,57.328c-0.8,0 -1.5,-0.5
-1.8,-1.2s-0.1,-1.5 0.4,-2.1c0.5,-0.5 1.4,-0.7 2.1,-0.4c0.7,0.3 1.2,1 1.2,1.8C66.5,56.528
65.6,57.328 64.6,57.328L64.6,57.328z"
      android:strokeWidth="1"
      android:strokeColor="#00000000" />
```


</vector>

``

Archivo: res\layout\activity_chat_ia.xml

``xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#F5F5F5"
    tools:context=".ui.chat.ChatIAActivity">

    <!-- Encabezado del chat -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="?attr/colorPrimary"
        android:padding="20dp"
        android:orientation="vertical"
        android:elevation="4dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="ðŸ’¬ Mi Asistente"
            android:textSize="24sp"
            android:textStyle="bold"
            android:textColor="@android:color/white"
            android:gravity="center"/>

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="PregÃºntame sobre tu inventario"
            android:textSize="14sp"
            android:textColor="#FFFFFF"
            android:gravity="center"
            android:layout_marginTop="4dp"/>
    </LinearLayout>

    <!-- Ãrea de conversaciÃ³n -->
```

```

<ScrollView
    android:id="@+id/scroll_chat"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:padding="16dp"
    android:background="@android:color/white">

    <TextView
        android:id="@+id/txt_chat_history"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="16sp"
        android:lineSpacingExtra="6dp"
        android:textColor="#000000"/>
</ScrollView>

<!-- Área de entrada de mensaje -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:padding="12dp"
    android:background="@android:color/white"
    android:elevation="8dp">

    <EditText
        android:id="@+id/edt_question"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:hint="Escribe tu pregunta aquí..."
        android:padding="16dp"
        android:textSize="16sp"
        android:textColor="#000000"
        android:textColorHint="#757575"
        android:background="@android:drawable/editbox_background"
        android:layout_marginEnd="8dp"/>

    <Button
        android:id="@+id/btn_send"
        android:layout_width="wrap_content"
        android:layout_height="56dp"
        android:text="Enviar"
        android:textSize="22sp"
        android:minWidth="64dp"
        android:paddingStart="16dp"
        android:paddingEnd="16dp"/>

```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
``
```

```
## Archivo: res\layout\activity_history.xml
```

```
``xml
```

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="@color/background">
```

```
<!-- Encabezado -->
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:background="@drawable/header_background"
    android:paddingStart="24dp"
    android:paddingEnd="24dp"
    android:paddingTop="48dp"
    android:paddingBottom="32dp"
    android:elevation="4dp"
    android:layout_marginBottom="16dp">
```

```
<TextView
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ“Š Mis Ventas"
    android:textSize="28sp"
    android:textStyle="bold"
    android:textColor="@color/white"
    android:gravity="center_horizontal"/>
```

```
<TextView
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="AquÃ puedes ver todo lo que has vendido"
    android:textSize="16sp"
    android:textColor="#E0E0E0"
```

```
        android:gravity="center_horizontal"
        android:layout_marginTop="4dp"/>
</LinearLayout>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:paddingHorizontal="16dp">
```

```
<!-- Selector de período -->
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Ver ventas de:"
    android:textSize="16sp"
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:layout_marginBottom="8dp"/>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:layout_marginBottom="16dp">
```

```
<Button
    android:id="@+id/btn_filter_week"
    android:layout_width="0dp"
    android:layout_height="52dp"
    android:layout_weight="1"
    android:text="ðŸ“Š... Esta Semana"
    android:textSize="11sp"
    android:textAllCaps="false"
    android:paddingTop="8dp"
    android:paddingBottom="8dp"
    android:layout_marginEnd="4dp"
    style="@style/Widget.Material3.Button.TonalButton"/>
```

```
<Button
    android:id="@+id/btn_filter_month"
    android:layout_width="0dp"
    android:layout_height="52dp"
    android:layout_weight="1"
    android:text="ðŸ“Š† Este Mes"
    android:textSize="11sp"
    android:textAllCaps="false"
```

```
    android:paddingTop="8dp"
    android:paddingBottom="8dp"
    android:layout_marginStart="4dp"
    android:layout_marginEnd="4dp"
    style="@style/Widget.Material3.Button.OutlinedButton"/>
```

```
<Button
    android:id="@+id/btn_filter_all"
    android:layout_width="0dp"
    android:layout_height="52dp"
    android:layout_weight="1"
    android:text="Filtrar Todo"
    android:textSize="11sp"
    android:textAllCaps="false"
    android:paddingTop="8dp"
    android:paddingBottom="8dp"
    android:layout_marginStart="4dp"
    style="@style/Widget.Material3.Button.OutlinedButton"/>
```

```
</LinearLayout>
```

```
<!-- Estadísticas -->
```

```
<TextView
    android:id="@+id/txt_period_label"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Esta semana:"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:layout_marginBottom="12dp"/>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:background="@color/surface"
    android:padding="16dp"
    android:layout_marginBottom="16dp"
    android:elevation="2dp">
```

```
<TextView
    android:id="@+id/txt_weekly_sales"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="0 ventas"
    android:textSize="18sp"
    android:textColor="@color/text_primary"
    android:textStyle="bold"
```

```

        android:layout_marginBottom="8dp"/>

<TextView
    android:id="@+id/txt_weekly_revenue"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ’° $0.00"
    android:textSize="22sp"
    android:textColor="@color/action_sale"
    android:textStyle="bold"
    android:layout_marginBottom="8dp"/>

<TextView
    android:id="@+id/txt_total_movements"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ’° 0 movimientos totales"
    android:textSize="14sp"
    android:textColor="@color/text_secondary"/>
</LinearLayout>

<!-- BotÃ³n de reseteo -->
<Button
    android:id="@+id/btn_reset_week"
    android:layout_width="match_parent"
    android:layout_height="60dp"
    android:text="ðŸ’° „ Empezar Nueva Semana"
    android:textSize="16sp"
    android:textStyle="bold"
    android:textAllCaps="false"
    android:textColor="@color/primary"
    android:layout_marginBottom="16dp"
    style="@style/Widget.Material3.Button.OutlinedButton"/>

<!-- Lista de movimientos -->
<TextView
    android:id="@+id/txt_history_label"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Historial completo:"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:layout_marginBottom="4dp"/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"

```

```
        android:text="ðŸŒ™ MantÃ©n presionado para editar o eliminar"
        android:textSize="14sp"
        android:textColor="@color/text_secondary"
        android:textStyle="italic"
        android:layout_marginBottom="12dp"/>
```

```
    <ListView
        android:id="@+id/list_movements"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:divider="@color/divider"
        android:dividerHeight="1dp"
        android:background="@color/surface"/>
</LinearLayout>
```

```
</LinearLayout>
```

```
``
```

Archivo: res\layout\activity_main.xml

```
``xml
```

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/background"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical">

        <!-- Encabezado atractivo -->
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:background="@drawable/header_background"
            android:paddingStart="24dp"
```

```
android:paddingEnd="24dp"
android:paddingTop="48dp"
android:paddingBottom="32dp"
android:orientation="vertical"
android:elevation="4dp">
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ“¡ Mi Inventario"
    android:textSize="32sp"
    android:textStyle="bold"
    android:textColor="@color/white"
    android:gravity="center_horizontal"/>
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Todos mis productos"
    android:textSize="16sp"
    android:textColor="#E0E0E0"
    android:gravity="center_horizontal"
    android:layout_marginTop="4dp"/>
```

```
</LinearLayout>
```

```
<!-- Barra de búsqueda flotante -->
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginHorizontal="16dp"
    android:layout_marginTop="-24dp"
    android:padding="4dp"
    android:background="@drawable/bg_search"
    android:elevation="8dp">
```

```
<EditText
    android:id="@+id/search_products"
    android:layout_width="0dp"
    android:layout_height="48dp"
    android:layout_weight="1"
    android:hint="ðŸ“¡ Buscar producto..."
    android:textSize="16sp"
    android:textColor="@color/text_primary"
    android:textColorHint="@color/text_secondary"
    android:background="@android:color/transparent"
    android:paddingStart="16dp"
    android:paddingEnd="16dp"
```



```
android:singleLine="true"
android:imeOptions="actionSearch"
android:inputType="text"/>
```

```
<ImageButton
    android:id="@+id/btn_clear_search"
    android:layout_width="48dp"
    android:layout_height="48dp"
    android:src="@android:drawable/ic_menu_close_clear_cancel"
    android:background="?attr/selectableItemBackgroundBorderless"
    android:contentDescription="Limpiar bÃ³squeda"
    app:tint="@color/text_secondary"
    android:visibility="gone"/>
</LinearLayout>
```

<!-- Lista de productos con padding -->

```
<ListView
    android:id="@+id/list_products"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="8dp"
    android:clipToPadding="false"
    android:divider="@android:color/transparent"
    android:dividerHeight="8dp"
    android:layout_marginTop="8dp"/>
</LinearLayout>
```

<!-- BotÃ³n flotante principal (FAB) -->

```
<com.google.android.material.floatingactionbutton.FloatingActionButton
    android:id="@+id/fab_menu"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="bottom|end"
    android:layout_margin="24dp"
    android:src="@android:drawable/ic_menu_add"
    android:contentDescription="MenÃº de opciones"
    app:tint="@color/white"
    app:backgroundTint="@color/action_add"
    app:fabSize="normal"/>
```

<!-- MenÃº desplegable de opciones (inicialmente oculto) -->

```
<LinearLayout
    android:id="@+id/fab_menu_layout"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="bottom|end"
    android:layout_marginBottom="96dp"
    android:layout_marginEnd="24dp"
```

```
android:orientation="vertical"
android:visibility="gone">
```

```
<!-- Opci3n: Pregntame Algo -->
```

```
<LinearLayout
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:layout_marginBottom="16dp"
    android:layout_gravity="end">
```

```
<TextView
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="3 Pregntame Algo"
    android:background="@drawable/bg_menu_label"
    android:backgroundTint="@color/action_chat"
    android:paddingHorizontal="16dp"
    android:paddingVertical="8dp"
    android:textSize="14sp"
    android:textColor="@color/white"
    android:textStyle="bold"
    android:elevation="4dp"
    android:layout_marginEnd="16dp"/>
```

```
<com.google.android.material.floatingactionbutton.FloatingActionButton
```

```
    android:id="@+id/fab_chat"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:src="@android:drawable/ic_menu_help"
    app:fabSize="mini"
    app:tint="@color/white"
    app:backgroundTint="@color/action_chat"/>
```

```
</LinearLayout>
```

```
<!-- Opci3n: Registrar Venta/Entrada -->
```

```
<LinearLayout
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:layout_marginBottom="16dp"
    android:layout_gravity="end">
```

```
<TextView
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
```

```

        android:text="ðŸ“ Registrar Venta/Entrada"
        android:background="@drawable/bg_menu_label"
        android:backgroundTint="@color/action_movement"
        android:paddingHorizontal="16dp"
        android:paddingVertical="8dp"
        android:textSize="14sp"
        android:textColor="@color/white"
        android:textStyle="bold"
        android:elevation="4dp"
        android:layout_marginEnd="16dp"/>

```

```

<com.google.android.material.floatingactionbutton.FloatingActionButton
    android:id="@+id/fab_movements"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:src="@android:drawable/ic_menu_edit"
    app:fabSize="mini"
    app:tint="@color/white"
    app:backgroundTint="@color/action_movement"/>
</LinearLayout>

```

<!-- OpciÃ³n: Ver Mis Ventas -->

```

<LinearLayout
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:layout_marginBottom="16dp"
    android:layout_gravity="end">

```

```

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="ðŸ“ Ñ Ver Mis Ventas"
    android:background="@drawable/bg_menu_label"
    android:backgroundTint="@color/action_sale"
    android:paddingHorizontal="16dp"
    android:paddingVertical="8dp"
    android:textSize="14sp"
    android:textColor="@color/white"
    android:textStyle="bold"
    android:elevation="4dp"
    android:layout_marginEnd="16dp"/>

```

```

<com.google.android.material.floatingactionbutton.FloatingActionButton
    android:id="@+id/fab_history"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"

```

```

        android:src="@android:drawable/ic_menu_view"
        app:fabSize="mini"
        app:tint="@color/white"
        app:backgroundTint="@color/action_sale"/>
</LinearLayout>

<!-- Opción: Ver Reportes -->
<LinearLayout
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:layout_marginBottom="16dp"
    android:layout_gravity="end">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Ver Reportes"
        android:background="@drawable/bg_menu_label"
        android:backgroundTint="@color/action_report"
        android:paddingHorizontal="16dp"
        android:paddingVertical="8dp"
        android:textSize="14sp"
        android:textColor="@color/white"
        android:textStyle="bold"
        android:elevation="4dp"
        android:layout_marginEnd="16dp"/>

    <com.google.android.material.floatingactionbutton.FloatingActionButton
        android:id="@+id/fab_reports"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:src="@android:drawable/ic_menu_sort_by_size"
        app:fabSize="mini"
        app:tint="@color/white"
        app:backgroundTint="@color/action_report"/>
</LinearLayout>

<!-- Opción: Agregar Producto -->
<LinearLayout
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:layout_gravity="end">

    <TextView

```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="• Agregar Producto"
        android:background="@drawable/bg_menu_label"
        android:backgroundTint="@color/action_add"
        android:paddingHorizontal="16dp"
        android:paddingVertical="8dp"
        android:textSize="14sp"
        android:textColor="@color/white"
        android:textStyle="bold"
        android:elevation="4dp"
        android:layout_marginEnd="16dp"/>

```

```

        <com.google.android.material.floatingactionbutton.FloatingActionButton
            android:id="@+id/fab_add_product"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:src="@android:drawable/ic_input_add"
            app:fabSize="mini"
            app:tint="@color/white"
            app:backgroundTint="@color/action_add"/>
    </LinearLayout>
</LinearLayout>

```

```

</androidx.coordinatorlayout.widget.CoordinatorLayout>

```

```

..

```

```

## Archivo: res\layout\activity_movements.xml

```

```

``xml

```

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="@color/background"
    tools:context=".ui.movements.MovementsActivity">

    <!-- Encabezado -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"

```

```
android:background="@drawable/header_background"
android:paddingStart="24dp"
android:paddingEnd="24dp"
android:paddingTop="48dp"
android:paddingBottom="32dp"
android:orientation="vertical"
android:elevation="4dp">
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ“” Registrar Movimiento"
    android:textSize="28sp"
    android:textStyle="bold"
    android:textColor="@color/white"
    android:gravity="center_horizontal"/>
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Anota cuando vendas, compres o uses productos"
    android:textSize="16sp"
    android:textColor="#E0E0E0"
    android:gravity="center_horizontal"
    android:layout_marginTop="4dp"/>
```

```
</LinearLayout>
```

```
<!-- Formulario -->
```

```
<com.google.android.material.card.MaterialCardView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="16dp"
    android:layout_marginTop="-24dp"
    app:cardCornerRadius="16dp"
    app:cardElevation="4dp"
    app:cardBackgroundColor="@color/surface">
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="24dp">
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="â “ Â¿ QuÃ© hiciste?"
    android:textSize="18sp"
```

```
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:paddingBottom="12dp"/>
```

```
<Spinner
    android:id="@+id/spinner_type"
    android:layout_width="match_parent"
    android:layout_height="60dp"
    android:background="@drawable/bg_search"
    android:padding="16dp"
    android:layout_marginBottom="24dp"/>
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="¿Qué producto?"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:paddingBottom="12dp"/>
```

```
<Spinner
    android:id="@+id/spinner_product"
    android:layout_width="match_parent"
    android:layout_height="60dp"
    android:background="@drawable/bg_search"
    android:padding="16dp"
    android:layout_marginBottom="24dp"/>
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="¿Cuántos?"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:paddingBottom="12dp"/>
```

```
<com.google.android.material.textfield.TextInputLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Escribe la cantidad"
    android:layout_marginBottom="24dp"
    style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox">
    <com.google.android.material.textfield.TextInputEditText
        android:id="@+id/edt_quantity"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```

        android:inputType="number"
        android:textSize="20sp"
        android:textColor="@color/text_primary"
        android:textStyle="bold"/>
</com.google.android.material.textfield.TextInputLayout>

<Button
    android:id="@+id/btn_register"
    android:layout_width="match_parent"
    android:layout_height="64dp"
    android:text="âœ… Registrar"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textAllCaps="false"
    app:cornerRadius="12dp"/>
</LinearLayout>
</com.google.android.material.card.MaterialCardView>

</LinearLayout>

..

## Archivo: res\layout\activity_product_detail.xml

``xml

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fillViewport="true"
    android:background="@color/background"
    tools:context=".ui.product.ProductDetailActivity">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <!-- Encabezado -->
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"

```



```
android:background="@drawable/header_background"
android:paddingStart="24dp"
android:paddingEnd="24dp"
android:paddingTop="48dp"
android:paddingBottom="32dp"
android:orientation="vertical"
android:elevation="4dp">
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ“ Mi Producto"
    android:textSize="28sp"
    android:textStyle="bold"
    android:textColor="@color/white"
    android:gravity="center_horizontal"/>
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Completa la informaci3n de tu producto"
    android:textSize="16sp"
    android:textColor="#E0E0E0"
    android:gravity="center_horizontal"
    android:layout_marginTop="4dp"/>
```

```
</LinearLayout>
```

```
<!-- Formulario con padding y fondo blanco -->
```

```
<com.google.android.material.card.MaterialCardView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="16dp"
    android:layout_marginTop="-24dp"
    app:cardCornerRadius="16dp"
    app:cardElevation="4dp"
    app:cardBackgroundColor="@color/surface">
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="24dp">
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ“! Â¿QuÃ© producto es?"
    android:textSize="16sp"
```

```
android:textStyle="bold"
android:textColor="@color/text_primary"
android:paddingBottom="8dp"/>
```

```
<com.google.android.material.textfield.TextInputLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Ejemplo: Elotes, Mayonesa, etc."
    android:layout_marginBottom="16dp"
    style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox">
    <com.google.android.material.textfield.TextInputEditText
        android:id="@+id/edt_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:textColor="@color/text_primary"
        android:textStyle="bold"/>
</com.google.android.material.textfield.TextInputLayout>
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ“ ÅŒCÃ³mo lo mides?"
    android:textSize="16sp"
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:paddingBottom="8dp"/>
```

```
<com.google.android.material.textfield.TextInputLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Ejemplo: piezas, kilos, litros, bolsas"
    android:layout_marginBottom="16dp"
    style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox">
    <com.google.android.material.textfield.TextInputEditText
        android:id="@+id/edt_unit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:textColor="@color/text_primary"
        android:textStyle="bold"/>
</com.google.android.material.textfield.TextInputLayout>
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ“Š ÅŒCuÃ¡nto tengo ahora?"
    android:textSize="16sp"
```

```
android:textStyle="bold"
android:textColor="@color/text_primary"
android:paddingBottom="8dp"/>
```

```
<com.google.android.material.textfield.TextInputLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Cantidad actual en tu inventario"
    android:layout_marginBottom="16dp"
    style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox">
    <com.google.android.material.textfield.TextInputEditText
        android:id="@+id/edt_stock"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="number"
        android:textSize="20sp"
        android:textColor="@color/text_primary"
        android:textStyle="bold"/>
    </com.google.android.material.textfield.TextInputLayout>
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="¿Cuánto ánimo necesito tener?"
    android:textSize="16sp"
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:paddingBottom="8dp"/>
```

```
<com.google.android.material.textfield.TextInputLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Te avisaré cuando llegues a este número"
    android:layout_marginBottom="16dp"
    style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox">
    <com.google.android.material.textfield.TextInputEditText
        android:id="@+id/edt_min_stock"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="number"
        android:textSize="20sp"
        android:textColor="@color/text_primary"
        android:textStyle="bold"/>
    </com.google.android.material.textfield.TextInputLayout>
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
```

```

        android:text="¿A cuánto lo vendo?"
        android:textSize="16sp"
        android:textStyle="bold"
        android:textColor="@color/text_primary"
        android:paddingBottom="8dp"/>

```

```

<com.google.android.material.textfield.TextInputLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Precio de venta por unidad"
    android:layout_marginBottom="24dp"
    style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox">
    <com.google.android.material.textfield.TextInputEditText
        android:id="@+id/edt_price"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="numberDecimal"
        android:textSize="20sp"
        android:textColor="@color/text_primary"
        android:textStyle="bold"/>
</com.google.android.material.textfield.TextInputLayout>

```

```

<Button
    android:id="@+id/btn_save"
    android:layout_width="match_parent"
    android:layout_height="64dp"
    android:text="Guardar Producto"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textAllCaps="false"
    android:layout_marginBottom="12dp"
    app:cornerRadius="12dp"/>

```

```

<Button
    android:id="@+id/btn_delete"
    android:layout_width="match_parent"
    android:layout_height="64dp"
    android:text="Eliminar Producto"
    android:textSize="16sp"
    android:textColor="@color/action_delete"
    android:textStyle="bold"
    android:textAllCaps="false"
    android:backgroundTint="#FFEB3B"
    style="@style/Widget.Material3.Button.OutlinedButton"
    app:cornerRadius="12dp"
    app:strokeColor="@color/action_delete"/>

```

```

</LinearLayout>

```

```

</com.google.android.material.card.MaterialCardView>

```

```
</LinearLayout>
</ScrollView>
```

```
``
```

```
## Archivo: res\layout\activity_reports.xml
```

```
``xml
```

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/background">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:paddingBottom="16dp">

        <!-- Encabezado -->
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:background="@drawable/header_background"
            android:paddingStart="24dp"
            android:paddingEnd="24dp"
            android:paddingTop="48dp"
            android:paddingBottom="32dp"
            android:elevation="4dp"
            android:layout_marginBottom="16dp">

            <TextView
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="ðŸ“Š Reportes de Ventas"
                android:textSize="28sp"
                android:textStyle="bold"
                android:textColor="@color/white"
                android:gravity="center_horizontal"/>

            <TextView
                android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:text="Ve qué productos se venden más"
        android:textSize="16sp"
        android:textColor="#E0E0E0"
        android:gravity="center_horizontal"
        android:layout_marginTop="4dp"/>
</LinearLayout>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:paddingHorizontal="16dp">
```

```
<!-- Selector de período -->
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Mostrar datos de:"
    android:textSize="16sp"
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:layout_marginBottom="8dp"/>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="16dp">
```

```
<Button
    android:id="@+id/btn_period_week"
    android:layout_width="0dp"
    android:layout_height="52dp"
    android:layout_weight="1"
    android:text="📅 ... Esta Semana"
    android:textSize="12sp"
    android:textAllCaps="false"
    android:layout_marginEnd="4dp"
    style="@style/Widget.Material3.Button.TonalButton"/>
```

```
<Button
    android:id="@+id/btn_period_all"
    android:layout_width="0dp"
    android:layout_height="52dp"
    android:layout_weight="1"
    android:text="📅 Todo el Tiempo"
    android:textSize="12sp"
```

```
        android:textAllCaps="false"
        android:layout_marginStart="4dp"
        style="@style/Widget.Material3.Button.OutlinedButton"/>
</LinearLayout>
```

```
<!-- T tulo de secci n -->
```

```
<TextView
    android:id="@+id/txt_period_title"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="     Productos m s vendidos (esta semana)"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:layout_marginBottom="12dp"/>
```

```
<!-- Lista de productos m s vendidos -->
```

```
<ListView
    android:id="@+id/list_top_products"
    android:layout_width="match_parent"
    android:layout_height="400dp"
    android:background="@color/surface"
    android:divider="@color/divider"
    android:dividerHeight="1dp"
    android:layout_marginBottom="16dp"/>
```

```
<!-- Estad sticas diarias -->
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="     Ventas de los  ltimos 7 d as"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:layout_marginBottom="12dp"/>
```

```
<ListView
    android:id="@+id/list_daily_stats"
    android:layout_width="match_parent"
    android:layout_height="300dp"
    android:background="@color/surface"
    android:divider="@color/divider"
    android:dividerHeight="1dp"
    android:layout_marginBottom="16dp"/>
```

```
<!-- Resumen general -->
```

```
<TextView
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
    android:text="ðŸ”Œ Resumen General"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textColor="@color/text_primary"
    android:layout_marginBottom="12dp"/>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:background="@color/surface"
    android:padding="16dp"
    android:elevation="2dp">
```

```
<TextView
    android:id="@+id/txt_total_sales"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ”Œ Total de ventas: 0"
    android:textSize="16sp"
    android:textColor="@color/text_primary"
    android:layout_marginBottom="8dp"/>
```

```
<TextView
    android:id="@+id/txt_total_revenue"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ”Œ Ingresos totales: $0.00"
    android:textSize="18sp"
    android:textColor="@color/action_sale"
    android:textStyle="bold"/>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
</ScrollView>
```

```
..
```

```
## Archivo: res\layout\item_daily_stat.xml
```

```
``xml
```

```
<?xml version="1.0" encoding="utf-8"?>
```



```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="16dp"
    android:background="?attr/selectableItemBackground">
```

```
<TextView
    android:id="@+id/txt_day"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ“ˆ ... Lunes 18 de Nov"
    android:textSize="16sp"
    android:textStyle="bold"
    android:textColor="#000000"
    android:layout_marginBottom="8dp"/>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">
```

```
<LinearLayout
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:orientation="vertical">
```

```
<TextView
    android:id="@+id/txt_sales_count"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="25 ventas"
    android:textSize="14sp"
    android:textColor="#757575"/>
```

```
<TextView
    android:id="@+id/txt_items_sold"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ“ˆ 75 productos"
    android:textSize="14sp"
    android:textColor="#757575"
    android:layout_marginTop="4dp"/>
```

```
</LinearLayout>
```

```
<TextView
    android:id="@+id/txt_day_revenue"
```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="$750.00"
        android:textSize="20sp"
        android:textStyle="bold"
        android:textColor="#4CAF50"
        android:gravity="end"/>
</LinearLayout>

```

```

</LinearLayout>

```

```

..

```

Archivo: res\layout\item_movement.xml

```

``xml

```

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:padding="16dp"
    android:background="?attr/selectableItemBackground">

    <!-- Icono del tipo de movimiento -->
    <TextView
        android:id="@+id/txt_movement_type"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="ðŸ’"
        android:textSize="32sp"
        android:layout_marginEnd="16dp"/>

    <!-- Informaci3n del movimiento -->
    <LinearLayout
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:orientation="vertical">

        <TextView
            android:id="@+id/txt_movement_product"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"

```

```
    android:text="Producto"
    android:textSize="16sp"
    android:textStyle="bold"
    android:textColor="#000000"/>
```

```
<TextView
    android:id="@+id/txt_movement_quantity"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Cantidad: 10"
    android:textSize="14sp"
    android:textColor="#424242"
    android:layout_marginTop="4dp"/>
```

```
<TextView
    android:id="@+id/txt_movement_date"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Hace 2 días"
    android:textSize="12sp"
    android:textColor="#757575"
    android:layout_marginTop="4dp"/>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
..
```

```
## Archivo: res\layout\item_product.xml
```

```
``xml
```

```
<?xml version="1.0" encoding="utf-8"?>
<com.google.android.material.card.MaterialCardView
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginHorizontal="16dp"
    android:layout_marginVertical="6dp"
    app:cardCornerRadius="16dp"
    app:cardElevation="2dp"
    app:cardBackgroundColor="@color/surface"
    app:strokeColor="@color/divider"
    app:strokeWidth="1dp">
```

```

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:padding="16dp"
    android:gravity="center_vertical">

    <!-- Indicador de estado con círculo coloreado -->
    <View
        android:id="@+id/indicator_color"
        android:layout_width="14dp"
        android:layout_height="14dp"
        android:layout_marginEnd="16dp"
        android:background="@drawable/circle_indicator"/>

    <LinearLayout
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:orientation="vertical">

        <!-- Nombre del producto -->
        <TextView
            android:id="@+id/txt_name"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Producto"
            android:textSize="18sp"
            android:textStyle="bold"
            android:textColor="@color/text_primary"/>

        <!-- Stock con íconos -->
        <TextView
            android:id="@+id/txt_stock"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="📦 Tengo: 50 | 📦 MÃnimo: 20"
            android:textSize="14sp"
            android:textColor="@color/text_secondary"
            android:layout_marginTop="4dp"/>
    </LinearLayout>

    <!-- Precio destacado -->
    <TextView
        android:id="@+id/txt_price"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"

```

```

        android:text="$15.00"
        android:textSize="20sp"
        android:textStyle="bold"
        android:textColor="@color/primary"/>
    </LinearLayout>

</com.google.android.material.card.MaterialCardView>

```

``

Archivo: res\layout\item_top_product.xml

``xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="16dp"
    android:background="?attr/selectableItemBackground">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center_vertical">

        <TextView
            android:id="@+id/txt_rank"
            android:layout_width="40dp"
            android:layout_height="40dp"
            android:text="1"
            android:textSize="20sp"
            android:textStyle="bold"
            android:textColor="#FFFFFF"
            android:background="@drawable/circle_rank"
            android:gravity="center"
            android:layout_marginEnd="12dp"/>

        <LinearLayout
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:orientation="vertical">

```

```

<TextView
    android:id="@+id/txt_product_name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Elotes asados"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textColor="#000000"/>

<TextView
    android:id="@+id/txt_sold_quantity"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ“¡ 125 piezas vendidas"
    android:textSize="14sp"
    android:textColor="#757575"
    android:layout_marginTop="4dp"/>

<TextView
    android:id="@+id/txt_revenue"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ’° $1,250.00 en ventas"
    android:textSize="14sp"
    android:textColor="#4CAF50"
    android:textStyle="bold"
    android:layout_marginTop="4dp"/>

<TextView
    android:id="@+id/txt_times_sold"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ðŸ”” , Vendido 45 veces"
    android:textSize="12sp"
    android:textColor="#9E9E9E"
    android:layout_marginTop="2dp"/>
</LinearLayout>
</LinearLayout>

<!-- Barra de progreso visual -->
<View
    android:id="@+id/progress_bar"
    android:layout_width="match_parent"
    android:layout_height="6dp"
    android:layout_marginTop="12dp"
    android:background="?attr/colorPrimary"
    android:alpha="0.3"/>

```

```
</LinearLayout>
```

```
``
```

```
## Archivo: res\menu\main_menu.xml
```

```
``xml
```

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
  <item
    android:id="@+id/menu_add_product"
    android:icon="@android:drawable/ic_input_add"
    android:title="âž• Agregar Producto"/>
  <item
    android:id="@+id/menu_history"
    android:icon="@android:drawable/ic_menu_view"
    android:title="ðŸ“Š Ver Mis Ventas"/>
  <item
    android:id="@+id/menu_movements"
    android:icon="@android:drawable/ic_menu_edit"
    android:title="ðŸ“Š Registrar Venta/Entrada"/>
  <item
    android:id="@+id/menu_chat"
    android:icon="@android:drawable/ic_menu_help"
    android:title="ðŸ“Š ħ PregÃ°ntame Algo"/>
</menu>
```

```
``
```

```
## Archivo: res\mipmap-anydpi-v26\ic_launcher.xml
```

```
``xml
```

```
<?xml version="1.0" encoding="utf-8"?>
<adaptive-icon xmlns:android="http://schemas.android.com/apk/res/android">
  <background android:drawable="@drawable/ic_launcher_background" />
  <foreground android:drawable="@drawable/ic_launcher_foreground" />
  <monochrome android:drawable="@drawable/ic_launcher_foreground" />
</adaptive-icon>
```

``

Archivo: res\mipmap-anydpi-v26\ic_launcher_round.xml

``xml

```
<?xml version="1.0" encoding="utf-8"?>
<adaptive-icon xmlns:android="http://schemas.android.com/apk/res/android">
  <background android:drawable="@drawable/ic_launcher_background" />
  <foreground android:drawable="@drawable/ic_launcher_foreground" />
  <monochrome android:drawable="@drawable/ic_launcher_foreground" />
</adaptive-icon>
```

``

Archivo: res\values\colors.xml

``xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
  <color name="black">#FF000000</color>
  <color name="white">#FFFFFFFF</color>

  <!-- Brand Colors -->
  <color name="primary">#673AB7</color> <!-- Deep Purple 500 -->
  <color name="primary_variant">#512DA8</color> <!-- Deep Purple 700 -->
  <color name="secondary">#00BCD4</color> <!-- Cyan 500 -->
  <color name="secondary_variant">#0097A7</color> <!-- Cyan 700 -->

  <!-- Action Colors -->
  <color name="action_add">#9C27B0</color> <!-- Purple -->
  <color name="action_sale">#4CAF50</color> <!-- Green -->
  <color name="action_movement">#FF9800</color> <!-- Orange -->
  <color name="action_report">#03A9F4</color> <!-- Light Blue -->
  <color name="action_chat">#2196F3</color> <!-- Blue -->
  <color name="action_delete">#F44336</color> <!-- Red -->

  <!-- UI Colors -->
  <color name="background">#F5F5F5</color>
  <color name="surface">#FFFFFF</color>
  <color name="text_primary">#212121</color>
  <color name="text_secondary">#757575</color>
  <color name="divider">#E0E0E0</color>
```



```
</resources>
```

```
``
```

```
## Archivo: res\values\strings.xml
```

```
``xml
```

```
<resources>
```

```
    <string name="app_name">Mi Negocio</string>
```

```
    <!-- MainActivity -->
```

```
    <string name="title_main">Mi Inventario</string>
```

```
    <string name="btn_add_product">+ Nuevo Producto</string>
```

```
    <string name="btn_chat">💬 Pregúntame</string>
```

```
    <string name="btn_movements">📈 Ventas y Entradas</string>
```

```
    <!-- ProductDetailActivity -->
```

```
    <string name="title_product_detail">Mi Producto</string>
```

```
    <string name="hint_name">¿Qué producto es?</string>
```

```
    <string name="hint_unit">¿Cómo lo mides? (piezas, kilos, litros)</string>
```

```
    <string name="hint_stock">¿Cuánto tengo ahora?</string>
```

```
    <string name="hint_min_stock">¿Cuánto mínimo necesito tener?</string>
```

```
    <string name="hint_price">¿A cuánto lo vendo?</string>
```

```
    <string name="btn_save">💾 Guardar</string>
```

```
    <string name="btn_delete">🗑 Eliminar</string>
```

```
    <!-- MovementsActivity -->
```

```
    <string name="title_movements">Registrar Venta o Entrada</string>
```

```
    <string name="label_type">¿Qué hiciste?</string>
```

```
    <string name="label_product">¿Qué producto?</string>
```

```
    <string name="hint_quantity">¿Cuántos?</string>
```

```
    <string name="btn_register">📝 Registrar</string>
```

```
    <!-- ChatIAActivity -->
```

```
    <string name="title_chat">💬 Mi Asistente</string>
```

```
    <string name="hint_question">Pregúntame lo que quieras!</string>
```

```
    <string name="btn_send">📤 Enviar</string>
```

```
    <!-- Movement Types -->
```

```
    <string-array name="movement_types">
```

```
        <item>🛒 Entrar mercancía (comprar o recibir)</item>
```

```
        <item>📦 Salir mercancía (usar o regalar)</item>
```

```
        <item>📉 Vender</item>
```

```
    </string-array>
```

```

<!-- Messages -->
<string name="msg_product_saved">âœ… Â¡Listo! Producto guardado</string>
<string name="msg_product_updated">âœ… Â¡Actualizado correctamente!</string>
<string name="msg_product_deleted">ðŸ—‘ï, Producto eliminado</string>
<string name="msg_movement_registered">âœ… Â¡Registrado! Tu inventario se
actualizÃ³</string>
<string name="msg_name_required">âšš ï, Por favor escribe el nombre del
producto</string>
<string name="msg_quantity_required">âšš ï, Â¿CuÃ¡ntos fueron? Escribe la
cantidad</string>
<string name="msg_no_products">ðŸ“¡ AÃºn no tienes productos. Â¡Agrega tu primer
producto!</string>

```

```

<!-- Notifications -->
<string name="notif_channel_name">Alertas de Stock</string>
<string name="notif_channel_desc">Notificaciones de inventario bajo</string>
<string name="notif_low_stock_title">âšš ï, Inventario bajo</string>
</resources>

```

``

Archivo: res\values\themes.xml

``xml

```

<resources xmlns:tools="http://schemas.android.com/tools">
  <!-- Base application theme. -->
  <style name="Base.Theme.ProyectoFinalAppMobile"
parent="Theme.Material3.DayNight.NoActionBar">
    <!-- Customize your light theme here. -->
    <item name="colorPrimary">@color/primary</item>
    <item name="colorSecondary">@color/secondary</item>

    <item name="android:statusBarColor">@color/primary_variant</item>
    <item name="android:windowBackground">@color/background</item>
  </style>

  <style name="Theme.ProyectoFinalAppMobile"
parent="Base.Theme.ProyectoFinalAppMobile" />
</resources>

```

``

Archivo: res\values-night\themes.xml

```
``xml
```

```
<resources xmlns:tools="http://schemas.android.com/tools">
  <!-- Base application theme. -->
  <style name="Base.Theme.ProyectoFinalAppMobile"
parent="Theme.Material3.DayNight.NoActionBar">
  <!-- Customize your dark theme here. -->
  <!-- <item name="colorPrimary">@color/my_dark_primary</item> -->
  </style>
</resources>
```

```
``
```

Archivo: res\xml\backup_rules.xml

```
``xml
```

```
<?xml version="1.0" encoding="utf-8"?><!--
  Sample backup rules file; uncomment and customize as necessary.
  See https://developer.android.com/guide/topics/data/autobackup
  for details.
  Note: This file is ignored for devices older than API 31
  See https://developer.android.com/about/versions/12/backup-restore
-->
<full-backup-content>
  <!--
  <include domain="sharedpref" path="."/>
  <exclude domain="sharedpref" path="device.xml"/>
-->
</full-backup-content>
```

```
``
```

Archivo: res\xml\data_extraction_rules.xml

```
``xml
```

```
<?xml version="1.0" encoding="utf-8"?><!--
  Sample data extraction rules file; uncomment and customize as necessary.
  See https://developer.android.com/about/versions/12/backup-restore#xml-changes
  for details.
-->
<data-extraction-rules>
```

```

<cloud-backup>
  <!-- TODO: Use <include> and <exclude> to control what is backed up.
  <include .../>
  <exclude .../>
  -->
</cloud-backup>
<!--
<device-transfer>
  <include .../>
  <exclude .../>
</device-transfer>
-->
</data-extraction-rules>
``

```

Archivo: AndroidManifest.xml

```

``xml

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools">

  <!-- Permisos para futura integración con IA (API REST) -->
  <uses-permission android:name="android.permission.INTERNET" />
  <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
  <uses-permission android:name="android.permission.POST_NOTIFICATIONS" />

  <application
    android:allowBackup="true"
    android:dataExtractionRules="@xml/data_extraction_rules"
    android:fullBackupContent="@xml/backup_rules"
    android:icon="@mipmap/ic_launcher"
    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportRtl="true"
    android:theme="@style/Theme.ProyectoFinalAppMobile">

    <!-- MainActivity - Pantalla principal con lista de productos -->
    <activity
      android:name=".MainActivity"
      android:exported="true">
      <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />

```

```

        </intent-filter>
    </activity>

    <!-- ProductDetailActivity - Alta/edici3n de productos -->
    <activity
        android:name=".ui.product.ProductDetailActivity"
        android:exported="false"
        android:parentActivityName=".MainActivity" />

    <!-- MovementsActivity - Registro de movimientos (entradas/salidas/ventas) -->
    <activity
        android:name=".ui.movements.MovementsActivity"
        android:exported="false"
        android:parentActivityName=".MainActivity" />

    <!-- ChatIAActivity - Asistente de IA -->
    <activity
        android:name=".ui.chat.ChatIAActivity"
        android:exported="false"
        android:parentActivityName=".MainActivity" />

    <!-- HistoryActivity - Ver historial de ventas y estad3sticas -->
    <activity
        android:name=".HistoryActivity"
        android:exported="false"
        android:parentActivityName=".MainActivity" />

    <!-- ReportsActivity - Ver reportes de productos m3is vendidos -->
    <activity
        android:name=".ReportsActivity"
        android:exported="false"
        android:parentActivityName=".MainActivity" />

    <!-- StockService - Servicio de verificaci3n de stock bajo -->
    <service
        android:name=".services.StockService"
        android:exported="false" />

</application>

</manifest>

```